

Initial EDA: Wildfire School Closures and Absenteeism by Reason

Setup

This EDA follows three questions:

1. Are wildfires a major driver of instructional disruption?
2. Is the closure burden distributed unevenly across counties, districts, and schools?
3. What do absenteeism-by-reason patterns look like in schools with wildfire closure burden?

The closure dataset uses `year` as the ending year of the academic year. For example, 2019 represents the 2018-19 academic year. The absenteeism analysis therefore uses CDE Absenteeism by Reason files for 2017-18 and 2018-19, matched to closure `year == 2018` and `year == 2019`. The 2019-20 absenteeism file is intentionally excluded because CDE marks it unavailable/not valid due to COVID-era disruption.

This EDA uses CDE Absenteeism by Reason data, not the separate chronic absenteeism rate files. The outcome measures below therefore describe absence days and absence reasons among eligible students; they should not be interpreted as chronic absenteeism rates.

The primary wildfire burden metric is **wildfire missed student-days**, calculated as wildfire closure days multiplied by school-year enrollment. This metric is used because it reflects both the duration of disruption and the number of students exposed. Raw wildfire days and wildfire share of total closure days remain useful supporting measures, but missed student-days is the main scale for comparing burden across counties, districts, schools, and matched absenteeism records.

`days_quarantine` and `days_safety_other` appear to be subcategories of student safety in the codebook, so they are not included in the main top-level closure-cause totals below to avoid double counting.

```

library(tidyverse)
library(janitor)
library(scales)

theme_set(theme_minimal(base_size = 12))

to_number <- function(x) {
  readr::parse_number(as.character(x), na = c("", "NA", "N/A", "NULL", "*"))
}

safe_divide <- function(num, den) {
  if_else(!is.na(den) & den > 0, num / den, NA_real_)
}

clean_cds_code <- function(x) {
  digits <- str_remove_all(coalesce(as.character(x), ""), "\\D")
  if_else(digits == "", NA_character_, str_pad(digits, width = 14, side = "left", pad = "0"))
}

pad_code <- function(x, width) {
  digits <- str_remove_all(coalesce(as.character(x), ""), "\\D")
  if_else(digits == "", NA_character_, str_pad(digits, width = width, side = "left", pad = "0"))
}

year_breaks <- function(x) {
  seq(min(x, na.rm = TRUE), max(x, na.rm = TRUE), by = 2)
}

analysis_years <- c(2018, 2019)
analysis_year_labels <- c("2017-18", "2018-19")

cause_palette <- c(
  "Wildfire" = "#C7432B",
  "Natural disasters/weather" = "#3F7D8C",
  "Infrastructure" = "#D0A03A",
  "Student safety" = "#5E6E46",
  "Other" = "#7D7A78"
)

burden_palette <- c(
  "No wildfire" = "#BFC3C7",
  "Affected: lower 75%" = "#E7A15A",

```

```
"Affected: top 25%" = "#C7432B"
)
```

Load closure data

```
data_file <- "Annual school closures and standardized test scores in California, 2003-2019.csv"
data_path_candidates <- c(
  file.path("../school_fire_data", data_file),
  file.path("school_fire_data", data_file)
)

data_path <- data_path_candidates[file.exists(data_path_candidates)][1]
if (is.na(data_path)) {
  stop("Could not find the closure CSV. Check the relative path from this Quarto file.")
}

df <- readr::read_csv(
  data_path,
  col_types = readr::cols(
    cds_code = readr::col_character(),
    .default = readr::col_guess()
  ),
  show_col_types = FALSE
) |>
  clean_names()

# The provided column listing uses days_wildfires, while the codebook uses
# Days.Wildfire. Standardize to singular so downstream code is consistent.
if ("days_wildfires" %in% names(df) && !"days_wildfire" %in% names(df)) {
  df <- df |>
    rename(days_wildfire = days_wildfires)
}

primary_cause_vars <- c(
  "days_wildfire",
  "days_natural_disasters_weather",
  "days_infrastructure",
  "days_student_safety",
  "days_other"
)
```

```

closure_numeric_vars <- c(
  primary_cause_vars,
  "days_quarantine",
  "days_safety_other",
  "days_total",
  "total_enrollment"
)

df <- df |>
  mutate(
    year = as.integer(year),
    closure_year = year,
    academic_year = case_when(
      !is.na(year) ~ str_c(year - 1, "-", str_sub(as.character(year), 3, 4)),
      TRUE ~ NA_character_
    ),
    county = str_squish(str_to_title(county)),
    district_name = str_squish(district_name),
    school_site = str_squish(school_site),
    cds_code_raw = cds_code,
    cds_code = clean_cds_code(cds_code),
    across(any_of(closure_numeric_vars), to_number),
    across(any_of(primary_cause_vars), ~ replace_na(.x, 0)),
    days_total = replace_na(days_total, 0),
    total_enrollment = replace_na(total_enrollment, 0),
    wildfire_student_days = days_wildfire * total_enrollment,
    wildfire_share_of_total = safe_divide(days_wildfire, days_total),
    any_wildfire = days_wildfire > 0
  )

cause_lookup <- tibble(
  cause = primary_cause_vars,
  cause_label = factor(
    c(
      "Wildfire",
      "Natural disasters/weather",
      "Infrastructure",
      "Student safety",
      "Other"
    ),
    levels = c(
      "Wildfire",

```

```

    "Natural disasters/weather",
    "Infrastructure",
    "Student safety",
    "Other"
  )
)
)

glimpse(df)

```

Rows: 203,636

Columns: 116

\$ cds_code	<chr> "01098846987523", "011001~
\$ year	<int> 2015, 2006, 2007, 2008, 2~
\$ county	<chr> "Alameda", "Alameda", "Al~
\$ district_name	<chr> NA, "Alameda County Offic~
\$ school_site	<chr> NA, "FAME Public Charter"~
\$ days_wildfire	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ days_natural_disasters_weather	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ days_infrastructure	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ days_student_safety	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ days_quarantine	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ days_safety_other	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ days_other	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ days_total	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ total_enrollment	<dbl> 0, 785, 1143, 1286, 1398,~
\$ percent_female_enrollment	<dbl> NA, 0.5031847, 0.5030621,~
\$ percent_white_enrollment	<dbl> NA, 0.28280255, 0.2852143~
\$ percent_minority_enrollment	<dbl> NA, 0.7171975, 0.7147857,~
\$ percent_asian_enrollment	<dbl> NA, 0.286624204, 0.307961~
\$ percent_hispanic_enrollment	<dbl> NA, 0.08535032, 0.0979877~
\$ percent_black_enrollment	<dbl> NA, 0.14267516, 0.1268591~
\$ percent_esl	<dbl> NA, 0.30828026, 0.4138232~
\$ percent_free_reduced	<dbl> NA, NA, NA, 0.7946578, 0.~
\$ percentage_standard_exceeded_3_math	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_3_math	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_and_above_3_math	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_nearly_met_3_math	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_not_met_3_math	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_exceeded_4_math	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_4_math	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_and_above_4_math	<dbl> NA, NA, NA, NA, NA, NA, N~

[illegible]

\$ percentage_standard_exceeded_5_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_5_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_and_above_5_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_nearly_met_5_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_not_met_5_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_exceeded_6_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_6_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_and_above_6_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_nearly_met_6_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_not_met_6_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_exceeded_7_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_7_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_and_above_7_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_nearly_met_7_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_not_met_7_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_exceeded_8_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_8_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_and_above_8_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_nearly_met_8_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_not_met_8_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_exceeded_11_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_11_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_and_above_11_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_nearly_met_11_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_not_met_11_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_exceeded_13_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_13_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_and_above_13_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_nearly_met_13_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_not_met_13_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ act_avg_score	<dbl> NA, NA, NA, 18.00, NA, NA~
\$ sat_avg_verbal	<dbl> NA, NA, NA, NA, NA, 561.8~
\$ sat_avg_math	<dbl> NA, NA, NA, NA, NA, 576.4~
\$ sat_avg_writing_new	<dbl> NA, NA, NA, NA, NA, 537.2~
\$ sat_avg_total_old	<dbl> NA, NA, NA, NA, NA, 1138.~
\$ sat_avg_total_new	<dbl> NA, NA, NA, NA, NA, 1675.~
\$ percent_ap_345	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ closure_year	<int> 2015, 2006, 2007, 2008, 2~
\$ academic_year	<chr> "2014-15", "2005-06", "20~
\$ cds_code_raw	<chr> "\\01098846987523", "\\01~
\$ wildfire_student_days	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ wildfire_share_of_total	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ any_wildfire	<lgl> FALSE, FALSE, FALSE, FALS~

Closure data quality checks

```
df |>
  mutate(
    primary_cause_sum = rowSums(pick(all_of(primary_cause_vars)), na.rm = TRUE),
    total_minus_primary_causes = days_total - primary_cause_sum
  ) |>
  count(total_minus_primary_causes, sort = TRUE) |>
  slice_head(n = 20)
```

```
# A tibble: 1 x 2
  total_minus_primary_causes      n
      <dbl>   <int>
1             0 203636
```

```
df |>
  mutate(
    student_safety_subtotal = coalesce(days_quarantine, 0) + coalesce(days_safety_other, 0),
    student_safety_difference = days_student_safety - student_safety_subtotal
  ) |>
  count(student_safety_difference, sort = TRUE) |>
  slice_head(n = 20)
```

```
# A tibble: 1 x 2
  student_safety_difference      n
      <dbl>   <int>
1             0 203636
```

```
df |>
  count(cds_code, closure_year, sort = TRUE) |>
  filter(n > 1) |>
  slice_head(n = 20)
```

```
# A tibble: 2 x 3
  cds_code      closure_year      n
  <chr>          <int> <int>
1 31103143130093      2006      2
2 33752423331063      2005      2
```

Step 1: Is wildfire a major driver of instructional disruption?

```
statewide_by_year <- df |>
  group_by(year) |>
  summarise(
    schools = n_distinct(cds_code),
    schools_with_any_closure = n_distinct(cds_code[days_total > 0]),
    total_enrollment = sum(total_enrollment, na.rm = TRUE),
    across(all_of(primary_cause_vars), ~ sum(.x, na.rm = TRUE)),
    days_total = sum(days_total, na.rm = TRUE),
    .groups = "drop"
  ) |>
  mutate(
    primary_cause_sum = rowSums(pick(all_of(primary_cause_vars)), na.rm = TRUE),
    wildfire_share_of_total_days = safe_divide(days_wildfire, days_total),
    wildfire_share_of_primary_causes = safe_divide(days_wildfire, primary_cause_sum),
    focus_year = year %in% analysis_years
  )

statewide_long <- statewide_by_year |>
  select(year, all_of(primary_cause_vars)) |>
  pivot_longer(
    cols = all_of(primary_cause_vars),
    names_to = "cause",
    values_to = "closure_days"
  ) |>
  left_join(cause_lookup, by = "cause")

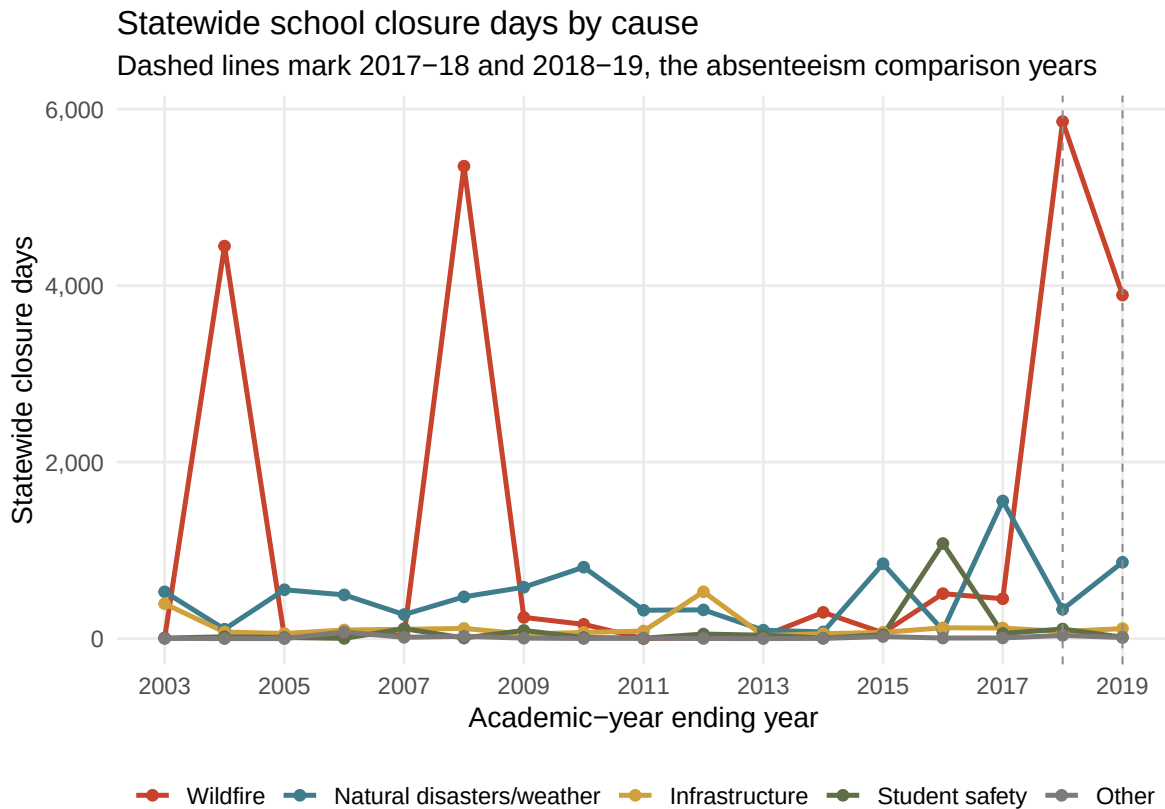
statewide_by_year |>
  arrange(desc(days_wildfire)) |>
  select(
    year,
    schools,
    days_wildfire,
    days_total,
    wildfire_share_of_total_days,
    wildfire_share_of_primary_causes
  ) |>
  slice_head(n = 10)
```

A tibble: 10 x 6

	year	schools	days_wildfire	days_total	wildfire_share_of_total_days
	<int>	<int>	<dbl>	<dbl>	<dbl>
1	2018	11677	5860	6415	0.913
2	2008	12249	5354	5974	0.896
3	2004	12072	4448	4648	0.957
4	2019	11581	3893	4898	0.795
5	2016	11572	509	1808	0.282
6	2017	11769	451	2194	0.206
7	2014	11555	297	434	0.684
8	2009	12185	240	970	0.247
9	2010	12109	161	1052	0.153
10	2015	13077	65	1049	0.0620

i 1 more variable: wildfire_share_of_primary_causes <dbl>

```
ggplot(
  statewide_long,
  aes(x = year, y = closure_days, color = cause_label, group = cause_label)
) +
  geom_line(linewidth = 1) +
  geom_point(size = 1.8) +
  geom_vline(xintercept = analysis_years, linetype = "dashed", color = "grey55", linewidth = 1) +
  scale_color_manual(values = cause_palette, name = NULL) +
  scale_x_continuous(breaks = year_breaks(statewide_long$year)) +
  scale_y_continuous(labels = label_comma()) +
  labs(
    title = "Statewide school closure days by cause",
    subtitle = "Dashed lines mark 2017-18 and 2018-19, the absenteeism comparison years",
    x = "Academic-year ending year",
    y = "Statewide closure days"
  ) +
  theme(
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  )
)
```



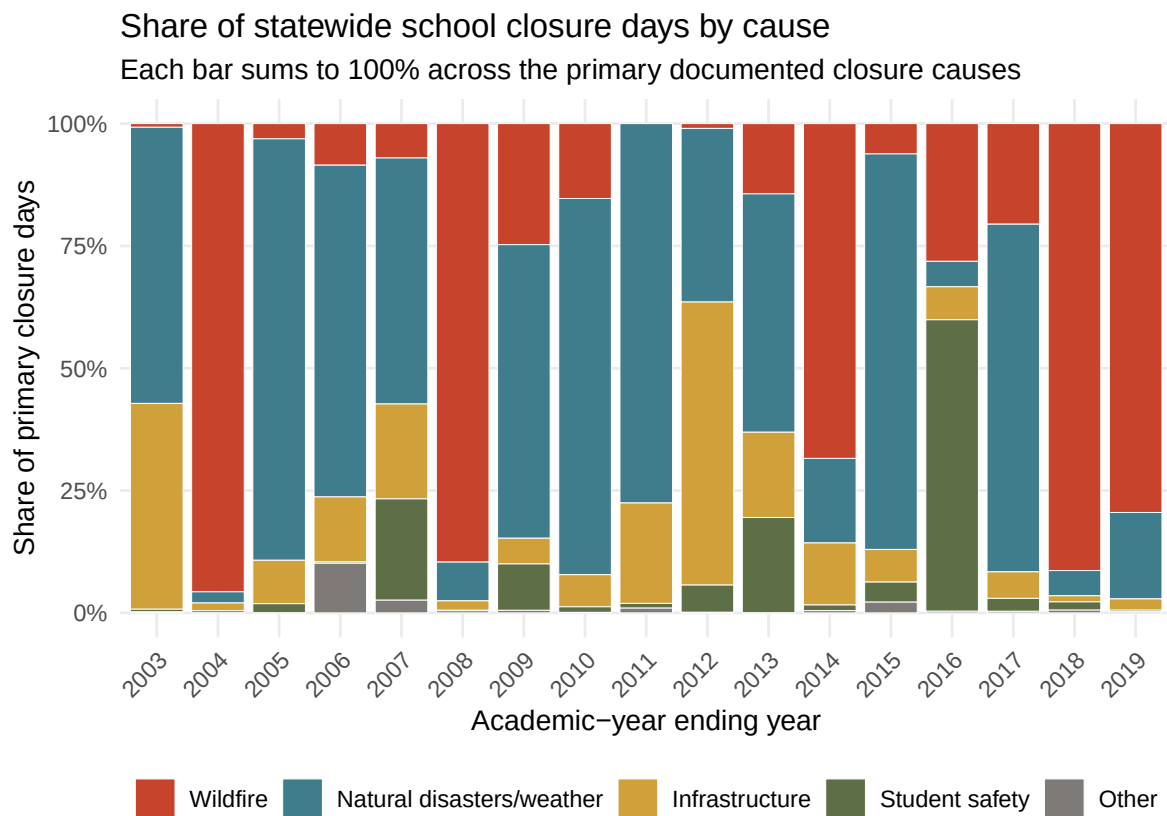
```
statewide_share <- statewide_long |>
  group_by(year) |>
  mutate(
    year_primary_total = sum(closure_days, na.rm = TRUE),
    share = safe_divide(closure_days, year_primary_total)
  ) |>
  ungroup() |>
  filter(!is.na(share))

ggplot(
  statewide_share,
  aes(x = factor(year), y = share, fill = cause_label)
) +
  geom_col(width = 0.86, color = "white", linewidth = 0.15) +
  scale_fill_manual(values = cause_palette, name = NULL) +
  scale_y_continuous(labels = label_percent()) +
  labs(
    title = "Share of statewide school closure days by cause",
```

```

  subtitle = "Each bar sums to 100% across the primary documented closure causes",
  x = "Academic-year ending year",
  y = "Share of primary closure days"
) +
theme(
  legend.position = "bottom",
  panel.grid.minor = element_blank(),
  axis.text.x = element_text(angle = 45, hjust = 1)
)

```



```

total_with_wildfire_component <- statewide_by_year |>
  transmute(
    year,
    Wildfire = days_wildfire,
    `Non-wildfire days` = pmax(days_total - days_wildfire, 0)
  ) |>
  pivot_longer(

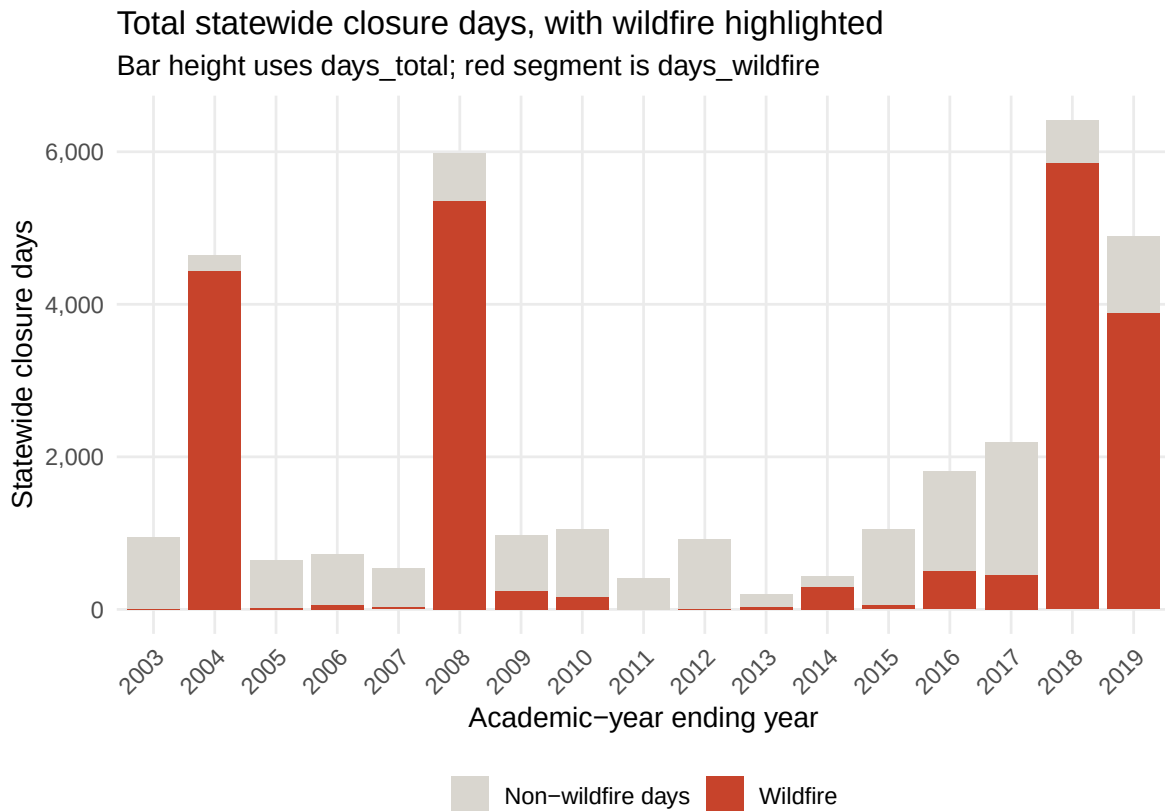
```

```

    cols = -year,
    names_to = "component",
    values_to = "closure_days"
  ) |>
  mutate(
    component = factor(component, levels = c("Non-wildfire days", "Wildfire"))
  )

ggplot(
  total_with_wildfire_component,
  aes(x = factor(year), y = closure_days, fill = component)
) +
  geom_col(width = 0.86) +
  scale_fill_manual(
    values = c(
      "Non-wildfire days" = "#D9D6CF",
      "Wildfire" = "#C7432B"
    ),
    name = NULL
  ) +
  scale_y_continuous(labels = label_comma()) +
  labs(
    title = "Total statewide closure days, with wildfire highlighted",
    subtitle = "Bar height uses days_total; red segment is days_wildfire",
    x = "Academic-year ending year",
    y = "Statewide closure days"
  ) +
  theme(
    legend.position = "bottom",
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

```



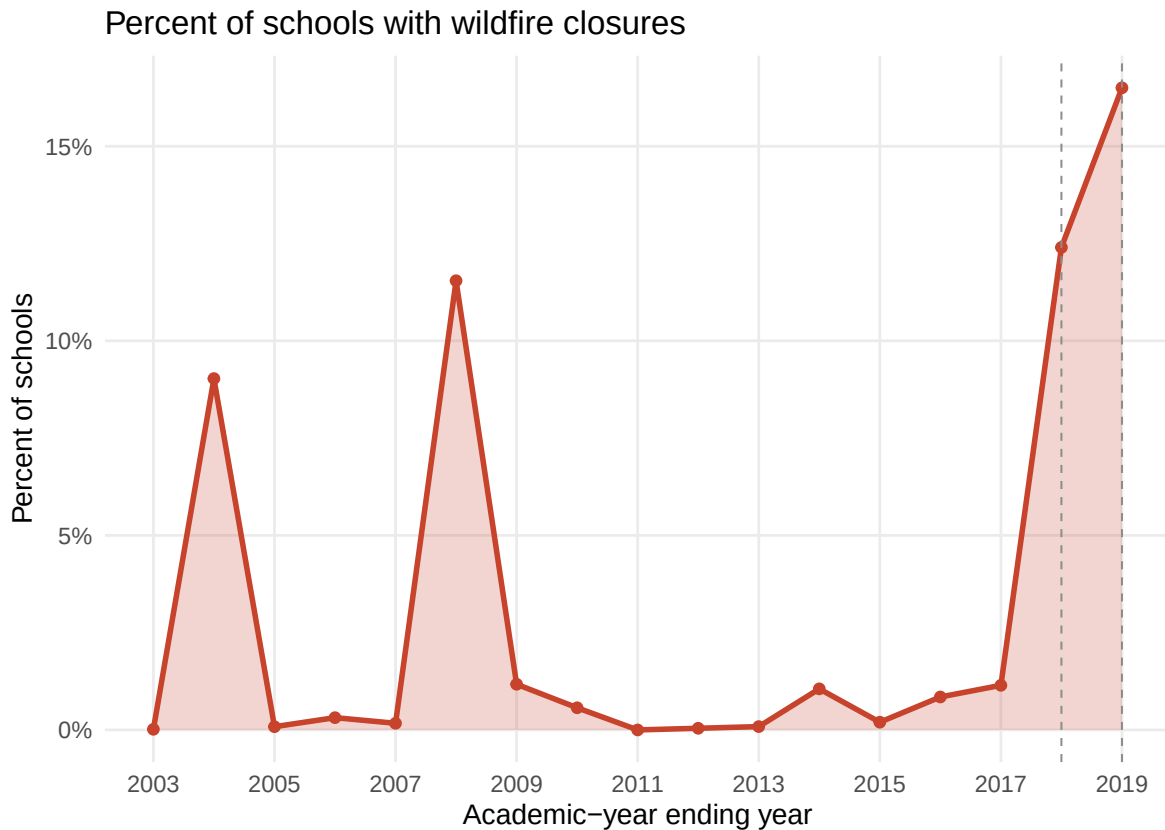
```
wildfire_reach_by_year <- df |>
  group_by(year) |>
  summarise(
    schools = n_distinct(cds_code),
    schools_with_wildfire_closure = n_distinct(cds_code[days_wildfire > 0]),
    pct_schools_with_wildfire_closure = safe_divide(schools_with_wildfire_closure, schools),
    wildfire_days = sum(days_wildfire, na.rm = TRUE),
    .groups = "drop"
  )

ggplot(
  wildfire_reach_by_year,
  aes(x = year, y = pct_schools_with_wildfire_closure)
) +
  geom_area(fill = "#C7432B", alpha = 0.22) +
  geom_line(color = "#C7432B", linewidth = 1.1) +
  geom_point(color = "#C7432B", size = 1.8) +
  geom_vline(xintercept = analysis_years, linetype = "dashed", color = "grey55", linewidth =
```

```

scale_x_continuous(breaks = year_breaks(wildfire_reach_by_year$year)) +
scale_y_continuous(labels = label_percent(), limits = c(0, NA)) +
labs(
  title = "Percent of schools with wildfire closures",
  x = "Academic-year ending year",
  y = "Percent of schools"
) +
theme(panel.grid.minor = element_blank())

```



Step 1 takeaway

Wildfire is a major driver of statewide instructional disruption, especially in peak closure years. The statewide rankings and cause-share plots show that 2018 and 2019 are central to the recent pre-pandemic wildfire story: wildfire closure days account for most statewide closure days in those years. This supports focusing the absenteeism comparison on the matched 2017-18 and 2018-19 school years.

Step 2: Is the burden uneven across counties, districts, and schools?

This section focuses on 2017-18 and 2018-19 to match the absenteeism files.

```
closure_window <- df |>
  filter(year %in% analysis_years)

county_window_burden <- closure_window |>
  group_by(county) |>
  summarise(
    school_year_rows = n(),
    schools = n_distinct(cds_code),
    districts = n_distinct(district_name),
    total_enrollment_school_years = sum(total_enrollment, na.rm = TRUE),
    wildfire_days = sum(days_wildfire, na.rm = TRUE),
    total_closure_days = sum(days_total, na.rm = TRUE),
    wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
    avg_wildfire_days_per_school_year = safe_divide(wildfire_days, school_year_rows),
    wildfire_share_of_total = safe_divide(wildfire_days, total_closure_days),
    .groups = "drop"
  ) |>
  filter(!is.na(county), county != "")

district_window_burden <- closure_window |>
  group_by(county, district_name) |>
  summarise(
    school_year_rows = n(),
    schools = n_distinct(cds_code),
    total_enrollment_school_years = sum(total_enrollment, na.rm = TRUE),
    wildfire_days = sum(days_wildfire, na.rm = TRUE),
    total_closure_days = sum(days_total, na.rm = TRUE),
    wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
    avg_wildfire_days_per_school_year = safe_divide(wildfire_days, school_year_rows),
    wildfire_share_of_total = safe_divide(wildfire_days, total_closure_days),
    .groups = "drop"
  ) |>
  filter(!is.na(district_name), district_name != "")

school_window_burden <- closure_window |>
  group_by(cds_code, county, district_name, school_site) |>
  summarise(
    school_year_rows = n(),
    years_observed = n_distinct(year),
```

```

total_enrollment_school_years = sum(total_enrollment, na.rm = TRUE),
wildfire_days = sum(days_wildfire, na.rm = TRUE),
total_closure_days = sum(days_total, na.rm = TRUE),
wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
avg_wildfire_days_per_school_year = safe_divide(wildfire_days, school_year_rows),
wildfire_share_of_total = safe_divide(wildfire_days, total_closure_days),
  .groups = "drop"
) |>
filter(!is.na(cds_code))

list(
  counties_with_wildfire = county_window_burden |> filter(wildfire_days > 0) |> nrow(),
  districts_with_wildfire = district_window_burden |> filter(wildfire_days > 0) |> nrow(),
  schools_with_wildfire = school_window_burden |> filter(wildfire_days > 0) |> nrow()
)

```

```

$counties_with_wildfire
[1] 36

```

```

$districts_with_wildfire
[1] 326

```

```

$schools_with_wildfire
[1] 2721

```

County burden rankings

```

top_n_counties <- 20

county_window_burden |>
  arrange(desc(wildfire_student_days)) |>
  select(
    county,
    schools,
    districts,
    wildfire_days,
    wildfire_student_days,
    avg_wildfire_days_per_school_year,
    wildfire_share_of_total
  )

```

```

) |>
slice_head(n = top_n_counties)

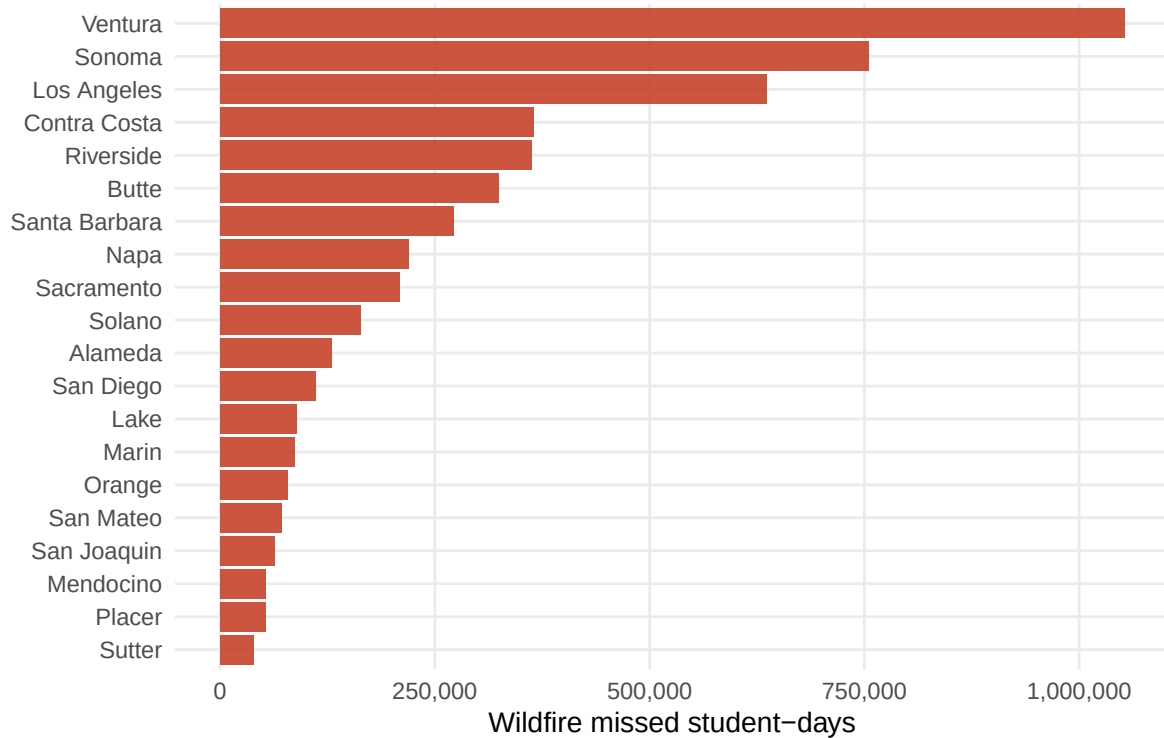
# A tibble: 20 x 7
  county      schools districts wildfire_days wildfire_student_days
  <chr>      <int>    <int>      <dbl>          <dbl>
1 Ventura      256      22      1540      1052437
2 Sonoma       244      43      1805      755475
3 Los Angeles  2508     100       879      635624
4 Contra Costa  322      22       596      364434
5 Riverside    576      27       371      362921
6 Butte        114      16       928      324612
7 Santa Barbara 142      27       570      271895
8 Napa          52       8       447      219721
9 Sacramento   425      17       305      209115
10 Solano       119       8       253      163051
11 Alameda     458      26       178      129436
12 San Diego   878      53       168      111207
13 Lake        46       8       344      88748
14 Marin       108      22       222      86526
15 Orange     693      34       107      78093
16 San Mateo   233      26       125      71285
17 San Joaquin 272      17       100      62862
18 Mendocino   77       15       181      53199
19 Placer      160      19       71      52751
20 Sutter       53      15       83      39253

# i 2 more variables: avg_wildfire_days_per_school_year <dbl>,
#   wildfire_share_of_total <dbl>

county_window_burden |>
  slice_max(wildfire_student_days, n = top_n_counties, with_ties = FALSE) |>
  ggplot(aes(x = wildfire_student_days, y = fct_reorder(county, wildfire_student_days))) +
  geom_col(fill = "#C7432B", alpha = 0.9) +
  scale_x_continuous(labels = label_comma()) +
  labs(
    title = "Counties with the largest wildfire missed student-days",
    subtitle = "2017-18 and 2018-19 only",
    x = "Wildfire missed student-days",
    y = NULL
  ) +
  theme(panel.grid.minor = element_blank())

```

Counties with the largest wildfire missed student-days 2017-18 and 2018-19 only



District burden rankings

```
top_n_districts <- 25

district_window_burden |>
  arrange(desc(wildfire_student_days)) |>
  select(
    county,
    district_name,
    schools,
    wildfire_days,
    wildfire_student_days,
    avg_wildfire_days_per_school_year,
    wildfire_share_of_total
  ) |>
  slice_head(n = top_n_districts)
```

```
# A tibble: 25 x 7
```

	county	district_name	schools	wildfire_days	wildfire_student_days
	<chr>	<chr>	<int>	<dbl>	<dbl>
1	Los Angeles	Los Angeles Unified	1017	789	572155
2	Ventura	Ventura Unified	28	390	251415
3	Napa	Napa Valley Unified	31	319	193670
4	Sonoma	Santa Rosa High	12	203	187754
5	Ventura	Oxnard Union High	13	114	175339
6	Ventura	Conejo Valley Unified	28	207	146813
7	Butte	Chico Unified	30	283	140817
8	Riverside	Corona-Norco Unified	54	139	139754
9	Ventura	Oxnard	21	160	132273
10	Riverside	Lake Elsinore Unified	26	115	106230

```
# i 15 more rows
```

```
# i 2 more variables: avg_wildfire_days_per_school_year <dbl>,
```

```
# wildfire_share_of_total <dbl>
```

```
district_window_burden |>
```

```
  mutate(district_label = str_c(str_trunc(district_name, 42), " (", county, ")")) |>
```

```
  slice_max(wildfire_student_days, n = 18, with_ties = FALSE) |>
```

```
  ggplot(aes(x = wildfire_student_days, y = fct_reorder(district_label, wildfire_student_days,
```

```
    geom_col(fill = "#B95A3C", alpha = 0.9) +
```

```
    scale_x_continuous(labels = label_comma()) +
```

```
    labs(
```

```
      title = "Districts with the largest wildfire missed student-days",
```

```
      subtitle = "2017-18 and 2018-19 only",
```

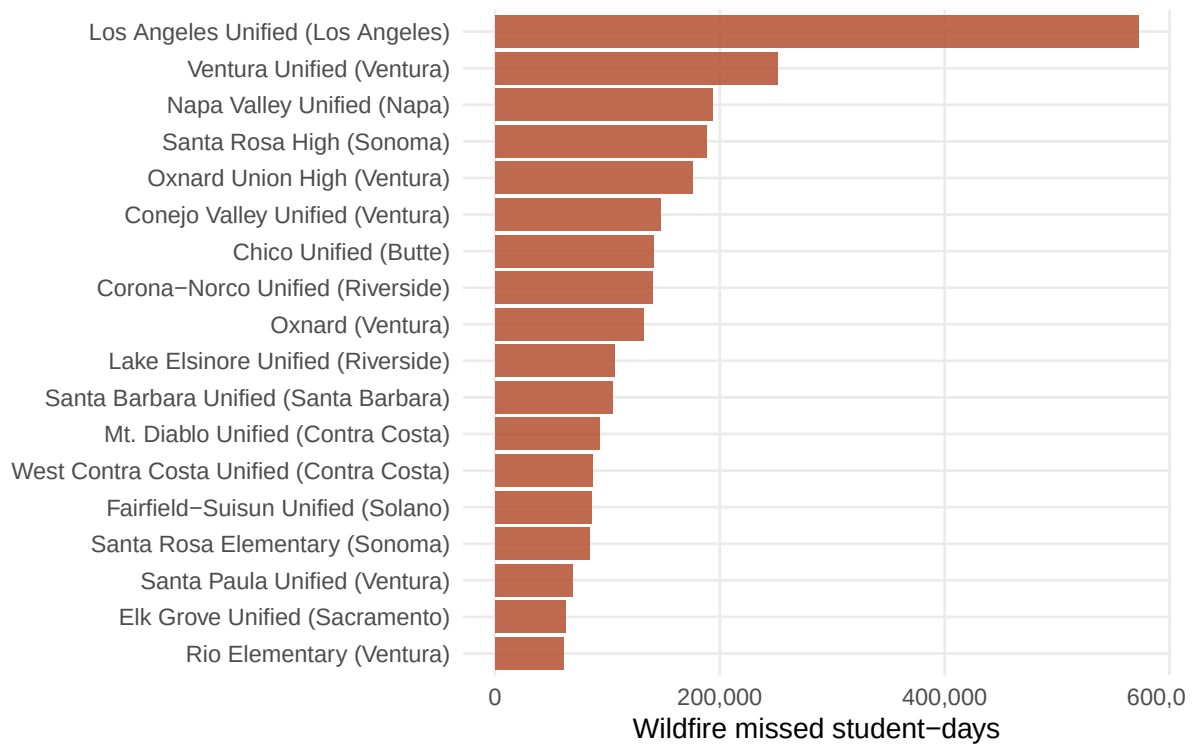
```
      x = "Wildfire missed student-days",
```

```
      y = NULL
```

```
    ) +
```

```
    theme(panel.grid.minor = element_blank())
```

Districts with the largest wildfire missed student-c
2017-18 and 2018-19 only



School burden rankings

```
top_n_schools <- 25

top_wildfire_schools <- school_window_burden |>
  arrange(desc(wildfire_student_days)) |>
  select(
    county,
    district_name,
    school_site,
    years_observed,
    total_enrollment_school_years,
    wildfire_days,
    wildfire_student_days,
    avg_wildfire_days_per_school_year,
    wildfire_share_of_total
  )
```

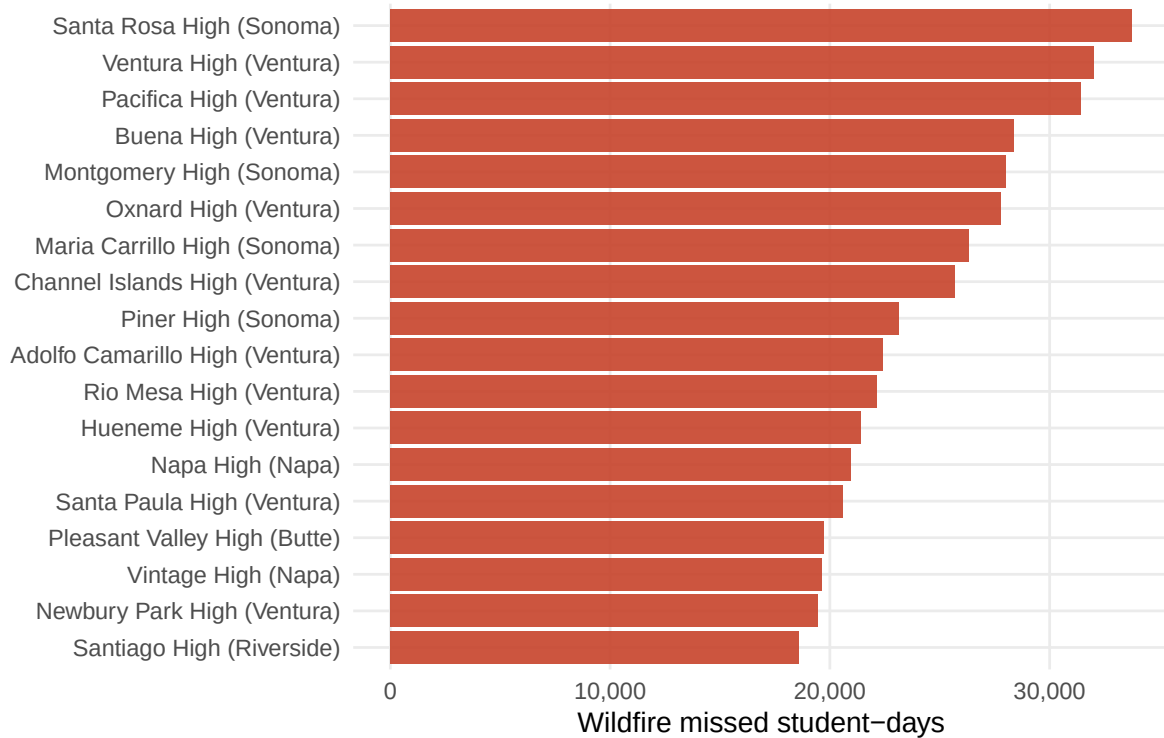
```
) |>
  slice_head(n = top_n_schools)

top_wildfire_schools
```

```
# A tibble: 25 x 9
  county district_name school_site years_observed total_enrollment_sch~1
  <chr>   <chr>          <chr>          <int>          <dbl>
1 Sonoma Santa Rosa High Santa Rosa H~      2          3939
2 Ventura Ventura Unified Ventura High      2          4207
3 Ventura Oxnard Union High Pacifica High      2          6252
4 Ventura Ventura Unified Buena High      2          3739
5 Sonoma Santa Rosa High Montgomery H~      2          3289
6 Ventura Oxnard Union High Oxnard High      2          5515
7 Sonoma Santa Rosa High Maria Carril~      2          3022
8 Ventura Oxnard Union High Channel Isla~      2          5155
9 Sonoma Santa Rosa High Piner High      2          2745
10 Ventura Oxnard Union High Adolfo Camar~      2          4470
# i 15 more rows
# i abbreviated name: 1: total_enrollment_school_years
# i 4 more variables: wildfire_days <dbl>, wildfire_student_days <dbl>,
#   avg_wildfire_days_per_school_year <dbl>, wildfire_share_of_total <dbl>
```

```
school_window_burden |>
  mutate(school_label = str_c(str_trunc(school_site, 42), " (", county, ")")) |>
  slice_max(wildfire_student_days, n = 18, with_ties = FALSE) |>
  ggplot(aes(x = wildfire_student_days, y = fct_reorder(school_label, wildfire_student_days))
  geom_col(fill = "#C7432B", alpha = 0.9) +
  scale_x_continuous(labels = label_comma()) +
  labs(
    title = "Schools with the largest wildfire missed student-days",
    subtitle = "2017-18 and 2018-19 only",
    x = "Wildfire missed student-days",
    y = NULL
  ) +
  theme(panel.grid.minor = element_blank())
```

Schools with the largest wildfire missed student-days 2017-18 and 2018-19 only



Concentration of wildfire burden

```
make_concentration <- function(data, unit_label) {
  data |>
    filter(wildfire_student_days > 0) |>
    arrange(desc(wildfire_student_days)) |>
    mutate(
      rank = row_number(),
      rank_share = rank / n(),
      cumulative_share = cumsum(wildfire_student_days) / sum(wildfire_student_days, na.rm = TRUE),
      unit_type = unit_label
    )
}

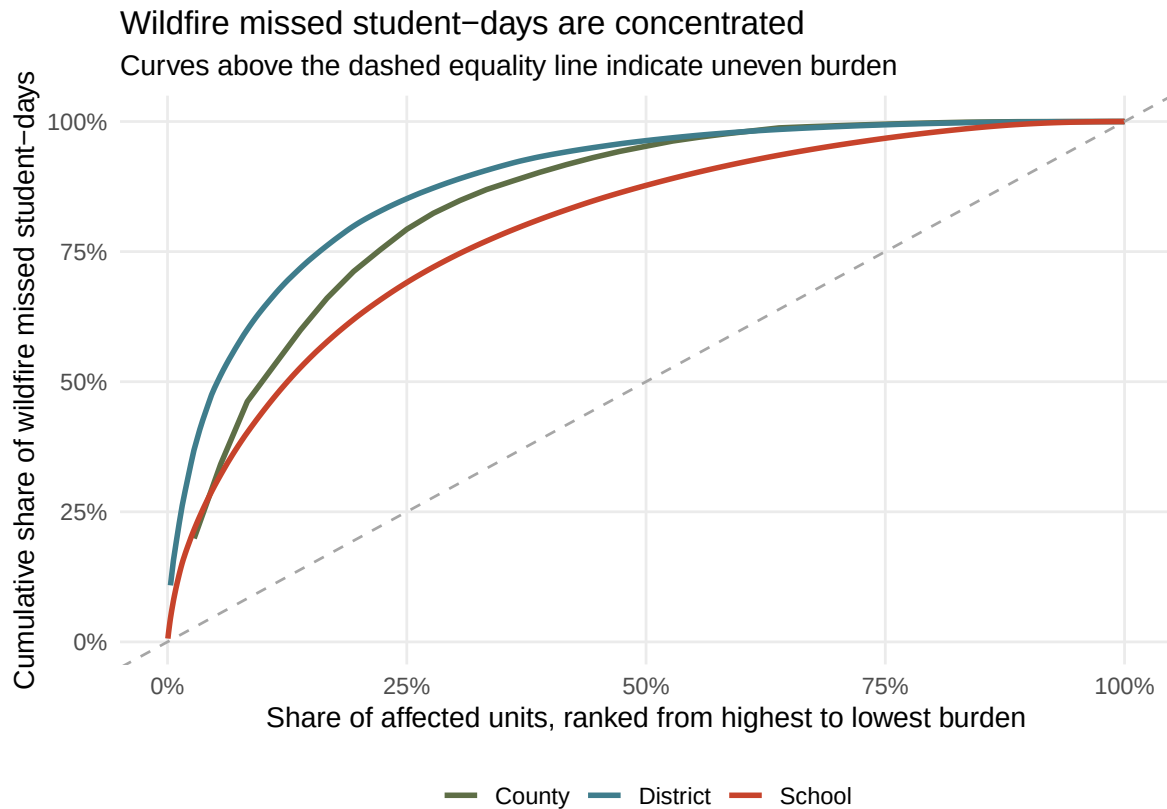
concentration_data <- bind_rows(
  make_concentration(school_window_burden, "School"),
```

```

make_concentration(district_window_burden, "District"),
make_concentration(county_window_burden, "County")
)

ggplot(
  concentration_data,
  aes(x = rank_share, y = cumulative_share, color = unit_type)
) +
  geom_line(linewidth = 1.1) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "grey65") +
  scale_x_continuous(labels = label_percent()) +
  scale_y_continuous(labels = label_percent()) +
  scale_color_manual(values = c("School" = "#C7432B", "District" = "#3F7D8C", "County" = "#5
  labs(
    title = "Wildfire missed student-days are concentrated",
    subtitle = "Curves above the dashed equality line indicate uneven burden",
    x = "Share of affected units, ranked from highest to lowest burden",
    y = "Cumulative share of wildfire missed student-days"
  ) +
  theme(
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  )

```



County maps

```
load_ca_counties <- function() {
  if (!requireNamespace("sf", quietly = TRUE)) {
    message("Package 'sf' is not installed; skipping map chunks.")
    return(NULL)
  }

  county_zip_url <- "https://www2.census.gov/geo/tiger/GENZ2018/shp/cb_2018_us_county_500k.zip"
  zip_path <- tempfile(fileext = ".zip")
  unzip_dir <- tempfile()
  dir.create(unzip_dir)

  tryCatch(
    {
      download.file(county_zip_url, zip_path, mode = "wb", quiet = TRUE)
    }
  )
}
```

```

unzip(zip_path, exdir = unzip_dir)
shp_path <- list.files(unzip_dir, pattern = "\\shp$", full.names = TRUE)[1]

sf::st_read(shp_path, quiet = TRUE) |>
  filter(STATEFP == "06") |>
  sf::st_transform(3310)
},
error = function(e) {
  message("County boundary download/read failed; skipping maps. Error: ", conditionMessage(e))
  NULL
}
)
}

ca_counties <- load_ca_counties()

county_map_data <- if (!is.null(ca_counties)) {
  ca_counties |>
  left_join(county_window_burden, by = c("NAME" = "county"))
} else {
  NULL
}

if (is.null(county_map_data)) {
  tibble(map_status = "County map data unavailable; non-map EDA sections still render.")
} else {
  county_map_data |>
  sf::st_drop_geometry() |>
  summarise(
    counties = n(),
    counties_matched_to_closure_data = sum(!is.na(wildfire_student_days)),
    counties_with_wildfire = sum(coalesce(wildfire_student_days, 0) > 0)
  )
}

```

	counties	counties_matched_to_closure_data	counties_with_wildfire
1	58	58	36

```

if (!is.null(county_map_data)) {
  p1 <- ggplot(county_map_data) +
    geom_sf(aes(fill = wildfire_student_days), color = "white", linewidth = 0.15) +

```

```

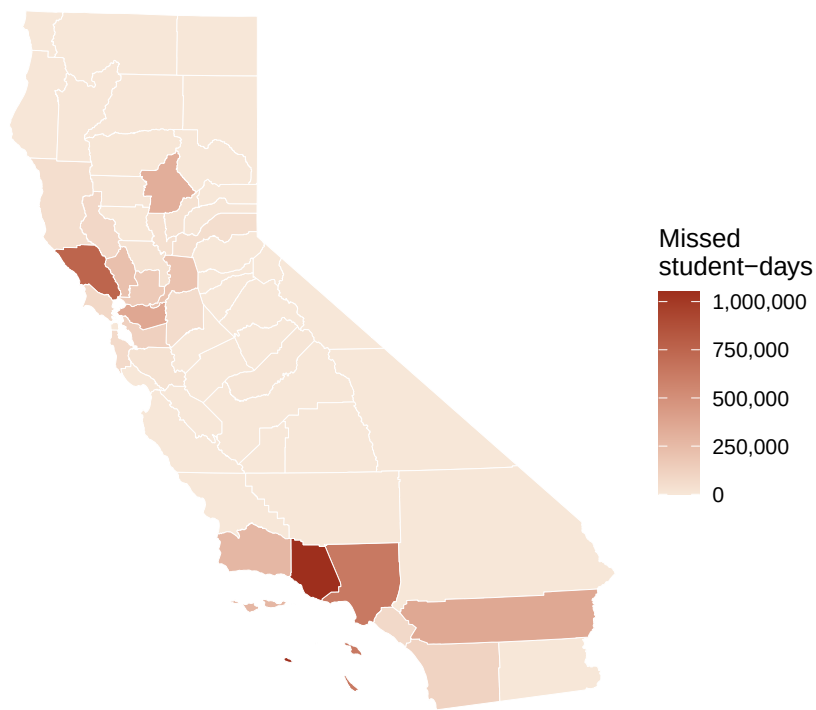
scale_fill_gradient(
  low = "#F7E7D8",
  high = "#9E2F1D",
  labels = label_comma(),
  na.value = "#F1F1F1",
  name = "Missed\nstudent-days"
) +
labs(
  title = "Wildfire missed student-days by county",
  subtitle = "2017-18 and 2018-19"
) +
theme_void() +
theme(legend.position = "right")

p2 <- ggplot(county_map_data) +
  geom_sf(aes(fill = avg_wildfire_days_per_school_year), color = "white", linewidth = 0.15)
scale_fill_gradient(
  low = "#F7E7D8",
  high = "#9E2F1D",
  labels = label_number(accuracy = 0.01),
  na.value = "#F1F1F1",
  name = "Avg days\nper school-year"
) +
labs(
  title = "Average wildfire closure days per school-year by county",
  subtitle = "2017-18 and 2018-19"
) +
theme_void() +
theme(legend.position = "right")

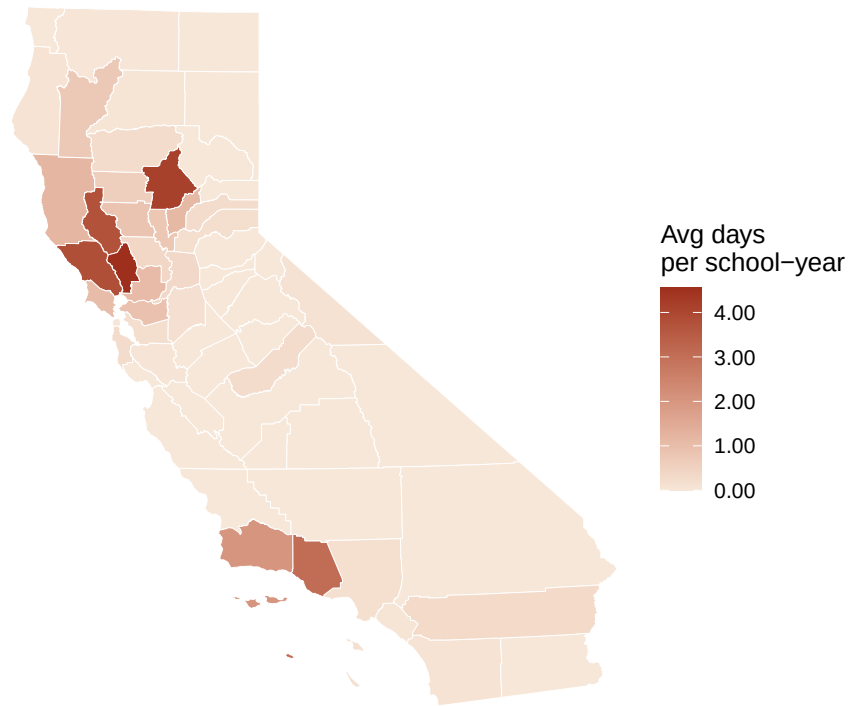
print(p1)
print(p2)
}

```

Wildfire missed student-days by county
2017-18 and 2018-19



Average wildfire closure days per school-year by county
2017–18 and 2018–19



Step 2 takeaway

Wildfire burden is widespread but uneven. In the 2017-18 and 2018-19 window, many counties, districts, and schools have some wildfire closure burden, but missed student-days are concentrated in a much smaller set of places. District counts can vary depending on whether districts are counted as county-district pairs or by district name alone; this notebook uses county-district pairs for the district burden tables because district names are not guaranteed to be unique statewide.

Step 3: What does absenteeism look like in affected schools?

Load CDE Absenteeism by Reason data

CDE file source: <https://www.cde.ca.gov/ds/ad/filesabr.asp>

```

absenteeism_col_names <- c(
  "academic_year",
  "aggregate_level",
  "county_code",
  "district_code",
  "school_code",
  "county_name",
  "district_name",
  "school_name",
  "charter",
  "dass",
  "reporting_category",
  "eligible_cumulative_enrollment",
  "count_students_one_or_more_absences",
  "average_days_absent",
  "total_days_absent",
  "excused_absences_percent",
  "unexcused_absences_percent",
  "out_of_school_suspension_absences_percent",
  "incomplete_independent_study_absences_percent",
  "excused_absences_count",
  "unexcused_absences_count",
  "out_of_school_suspension_absences_count",
  "incomplete_independent_study_absences_count"
)

absenteeism_numeric_vars <- c(
  "eligible_cumulative_enrollment",
  "count_students_one_or_more_absences",
  "average_days_absent",
  "total_days_absent",
  "excused_absences_percent",
  "unexcused_absences_percent",
  "out_of_school_suspension_absences_percent",
  "incomplete_independent_study_absences_percent",
  "excused_absences_count",
  "unexcused_absences_count",
  "out_of_school_suspension_absences_count",
  "incomplete_independent_study_absences_count"
)

absence_reason_count_vars <- c(

```

```

    "excused_absences_count",
    "unexcused_absences_count",
    "out_of_school_suspension_absences_count",
    "incomplete_independent_study_absences_count"
  )

read_absenteeism_file <- function(url, closure_year) {
  readr::read_tsv(
    url,
    skip = 1,
    col_names = absenteeism_col_names,
    col_types = readr::cols(.default = readr::col_character()),
    na = c("", "NA")
  ) |>
  clean_names() |>
  mutate(
    closure_year = closure_year,
    source_url = url
  )
}

absenteeism_sources <- tibble(
  academic_year = analysis_year_labels,
  closure_year = analysis_years,
  url = c(
    "https://www3.cde.ca.gov/demo-downloads/attendance/absenteeismreason18.txt",
    "https://www3.cde.ca.gov/demo-downloads/attendance/absenteeismreason19.txt"
  )
)

absenteeism_raw <- absenteeism_sources |>
  mutate(data = map2(url, closure_year, read_absenteeism_file)) |>
  select(data) |>
  unnest(cols = data)

absenteeism_schools <- absenteeism_raw |>
  filter(
    aggregate_level == "S",
    reporting_category == "TA"
  ) |>
  mutate(
    suppressed_numeric_fields = rowSums(across(all_of(absenteeism_numeric_vars), ~ .x == "*"))
  )

```

```

county_code_clean = pad_code(county_code, 2),
district_code_clean = pad_code(district_code, 5),
school_code_clean = pad_code(school_code, 7),
cds_code = if_else(
  !is.na(county_code_clean) & !is.na(district_code_clean) & !is.na(school_code_clean),
  str_c(county_code_clean, district_code_clean, school_code_clean),
  NA_character_
),
across(all_of(absenteeism_numeric_vars), to_number),
county_name = str_squish(str_to_title(county_name)),
district_name = str_squish(district_name),
school_name = str_squish(school_name),
absence_reason_count_sum = rowSums(pick(all_of(absence_reason_count_vars)), na.rm = TRUE),
absence_reason_count_difference = total_days_absent - absence_reason_count_sum,
share_students_with_absence = safe_divide(
  count_students_one_or_more_absences,
  eligible_cumulative_enrollment
),
avg_days_absent_among_absent = average_days_absent,
avg_days_absent_per_eligible = safe_divide(
  total_days_absent,
  eligible_cumulative_enrollment
),
excused_absence_share = safe_divide(excused_absences_count, total_days_absent),
unexcused_absence_share = safe_divide(unexcused_absences_count, total_days_absent),
suspension_absence_share = safe_divide(out_of_school_suspension_absences_count, total_days_absent),
incomplete_independent_study_absence_share = safe_divide(
  incomplete_independent_study_absences_count,
  total_days_absent
)
)
)

glimpse(absenteeism_schools)

```

Rows: 20,227

Columns: 39

\$ academic_year
18", "2~

<chr> "2017-18", "2017-

\$ aggregate_level

<chr> "S", "S", "S", "S", "S", ~

\$ county_code

<chr> "01", "01", "01", "01", ~

\$ district_code

<chr> "10017", "10017", "10017~

\$ school_code

<chr> "0130419", "0130401", "0~

\$ county_name	<chr> "Alameda", "Alameda", "A~
\$ district_name	<chr> "Alameda County Office o~
\$ school_name	<chr> "Alameda County Communit~
\$ charter	<chr> "No", "No", "Yes", "Yes"~
\$ dass	<chr> "Yes", "Yes", "No", "No"~
\$ reporting_category	<chr> "TA", "TA", "TA", "TA", ~
\$ eligible_cumulative_enrollment	<dbl> 304, 166, 270, 121, 630, ~
\$ count_students_one_or_more_absences	<dbl> 287, 109, 250, 8, 593, 4~
\$ average_days_absent	<dbl> 33.7, 4.8, 10.7, 1.9, 9.~
\$ total_days_absent	<dbl> 9675.0, 526.0, 2678.0, 1~
\$ excused_absences_percent	<dbl> 13.5, 61.4, 30.9, 0.0, 4~
\$ unexcused_absences_percent	<dbl> 31.2, 36.7, 69.0, 0.0, 5~
\$ out_of_school_suspension_absences_percent	<dbl> 0.4, 1.9, 0.1, 0.0, 0.7, ~
\$ incomplete_independent_study_absences_percent	<dbl> 54.8, 0.0, 0.0, 100.0, 0~
\$ excused_absences_count	<dbl> 1309.0, 323.0, 827.0, 0.~
\$ unexcused_absences_count	<dbl> 3019.0, 193.0, 1848.0, 0~
\$ out_of_school_suspension_absences_count	<dbl> 42.0, 10.0, 3.0, 0.0, 43~
\$ incomplete_independent_study_absences_count	<dbl> 5305.0, 0.0, 0.0, 15.0, ~
\$ closure_year	<dbl> 2018, 2018, 2018, 2018, ~
\$ source_url	<chr> "https://www3.cde.ca.gov~
\$ suppressed_numeric_fields	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, ~
\$ county_code_clean	<chr> "01", "01", "01", "01", ~
\$ district_code_clean	<chr> "10017", "10017", "10017~
\$ school_code_clean	<chr> "0130419", "0130401", "0~
\$ cds_code	<chr> "01100170130419", "01100~
\$ absence_reason_count_sum	<dbl> 9675.0, 526.0, 2678.0, 1~
\$ absence_reason_count_difference	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, ~
\$ share_students_with_absence	<dbl> 0.9440789, 0.6566265, 0.~
\$ avg_days_absent_among_absent	<dbl> 33.7, 4.8, 10.7, 1.9, 9.~
\$ avg_days_absent_per_eligible	<dbl> 31.8256579, 3.1686747, 9~
\$ excused_absence_share	<dbl> 0.1352972, 0.6140684, 0.~
\$ unexcused_absence_share	<dbl> 0.3120413, 0.3669202, 0.~
\$ suspension_absence_share	<dbl> 0.0043410853, 0.01901140~
\$ incomplete_independent_study_absence_share	<dbl> 0.548320413, 0.000000000~

Absenteeism data quality checks

```
absenteeism_schools |>
  summarise(
    rows = n(),
    school_years = n_distinct(str_c(cds_code, closure_year)),
```

```

schools = n_distinct(cds_code),
rows_with_suppressed_values = sum(suppressed_numeric_fields > 0, na.rm = TRUE),
rows_missing_cds_code = sum(is.na(cds_code))
)

```

```

# A tibble: 1 x 5
  rows school_years schools rows_with_suppressed_values rows_missing_cds_code
<int>      <int>   <int>                <int>                <int>
1 20227        20227  10336                309                0

```

```

absenteeism_schools |>
  count(cds_code, closure_year, sort = TRUE) |>
  filter(n > 1) |>
  slice_head(n = 20)

```

```

# A tibble: 0 x 3
# i 3 variables: cds_code <chr>, closure_year <dbl>, n <int>

```

```

absenteeism_schools |>
  summarise(
    across(
      all_of(absenteeism_numeric_vars),
      ~ mean(is.na(.x)),
      .names = "missing_share_{.col}"
    )
  ) |>
  pivot_longer(everything(), names_to = "field", values_to = "missing_share") |>
  arrange(desc(missing_share)) |>
  slice_head(n = 20)

```

```

# A tibble: 12 x 2
  field missing_share
<chr>      <dbl>
1 missing_share_eligible_cumulative_enrollment 0.0153
2 missing_share_count_students_one_or_more_absences 0.0153
3 missing_share_average_days_absent 0.0153
4 missing_share_total_days_absent 0.0153
5 missing_share_excused_absences_percent 0.0153
6 missing_share_unexcused_absences_percent 0.0153
7 missing_share_out_of_school_suspension_absences_percent 0.0153

```

```

8 missing_share_incomplete_independent_study_absences_percent      0.0153
9 missing_share_excused_absences_count                             0.0153
10 missing_share_unexcused_absences_count                           0.0153
11 missing_share_out_of_school_suspension_absences_count            0.0153
12 missing_share_incomplete_independent_study_absences_count       0.0153

```

```

absenteeism_schools |>
  mutate(abs_reason_abs_diff = abs(absence_reason_count_difference)) |>
  summarise(
    rows = n(),
    median_abs_difference = median(abs_reason_abs_diff, na.rm = TRUE),
    p95_abs_difference = quantile(abs_reason_abs_diff, 0.95, na.rm = TRUE),
    max_abs_difference = max(abs_reason_abs_diff, na.rm = TRUE)
  )

```

```

# A tibble: 1 x 4
  rows median_abs_difference p95_abs_difference max_abs_difference
<int>          <dbl>          <dbl>          <dbl>
1 20227             0             0             0.200

```

Join closure school-years to absenteeism school-years

```

closure_school_year <- closure_window |>
  group_by(cds_code, closure_year) |>
  summarise(
    academic_year = first(academic_year),
    county = first(county),
    district_name = first(district_name),
    school_site = first(school_site),
    total_enrollment = max(total_enrollment, na.rm = TRUE),
    days_wildfire = sum(days_wildfire, na.rm = TRUE),
    days_total = sum(days_total, na.rm = TRUE),
    wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
    wildfire_share_of_total = safe_divide(days_wildfire, days_total),
    any_wildfire = any(days_wildfire > 0, na.rm = TRUE),
    .groups = "drop"
  ) |>
  mutate(total_enrollment = if_else(is.infinite(total_enrollment), NA_real_, total_enrollment))

closure_absenteeism_joined <- closure_school_year |>

```

```

left_join(
  absenteeism_schools |>
  select(
    cds_code,
    closure_year,
    absenteeism_academic_year = academic_year,
    county_name_absenteeism = county_name,
    district_name_absenteeism = district_name,
    school_name_absenteeism = school_name,
    charter,
    dass,
    eligible_cumulative_enrollment,
    count_students_one_or_more_absences,
    average_days_absent,
    total_days_absent,
    all_of(absence_reason_count_vars),
    share_students_with_absence,
    avg_days_absent_among_absent,
    avg_days_absent_per_eligible,
    excused_absence_share,
    unexcused_absence_share,
    suspension_absence_share,
    incomplete_independent_study_absence_share,
    suppressed_numeric_fields
  ),
  by = c("cds_code", "closure_year")
)

affected_cutoff <- closure_absenteeism_joined |>
  filter(wildfire_student_days > 0, !is.na(avg_days_absent_per_eligible)) |>
  summarise(cutoff = quantile(wildfire_student_days, 0.75, na.rm = TRUE)) |>
  pull(cutoff)

if (length(affected_cutoff) == 0 || is.na(affected_cutoff)) {
  affected_cutoff <- Inf
}

closure_absenteeism_joined <- closure_absenteeism_joined |>
  mutate(
    burden_group = case_when(
      wildfire_student_days <= 0 ~ "No wildfire",
      wildfire_student_days >= affected_cutoff ~ "Affected: top 25%",

```

```

    TRUE ~ "Affected: lower 75%"
  ),
  burden_group = factor(
    burden_group,
    levels = c("No wildfire", "Affected: lower 75%", "Affected: top 25%")
  )
)

join_coverage <- tibble(
  metric = c(
    "closure_school_year_rows",
    "closure_school_year_rows_matched_to_absenteeism",
    "closure_school_year_match_rate",
    "unmatched_closure_school_year_rows",
    "absenteeism_school_year_rows_not_in_closure_data"
  ),
  value = c(
    nrow(closure_school_year),
    sum(!is.na(closure_absenteeism_joined$eligible_cumulative_enrollment)),
    safe_divide(
      sum(!is.na(closure_absenteeism_joined$eligible_cumulative_enrollment)),
      nrow(closure_school_year)
    ),
    nrow(anti_join(closure_school_year, absenteeism_schools, by = c("cds_code", "closure_year")),
    nrow(anti_join(absenteeism_schools, closure_school_year, by = c("cds_code", "closure_year"))
  )
)

join_coverage

```

```
# A tibble: 5 x 2
```

	metric <chr>	value <dbl>
1	closure_school_year_rows	23258
2	closure_school_year_rows_matched_to_absenteeism	19867
3	closure_school_year_match_rate	0.854
4	unmatched_closure_school_year_rows	3158
5	absenteeism_school_year_rows_not_in_closure_data	127

The join covers most closure school-year rows, but not all of them. Unmatched closure rows remain in the closure data but do not contribute to absenteeism summaries, and absenteeism

rows not found in the closure data are excluded from the closure-centered join. The unmatched examples below suggest that some non-matches are likely institutional or nonstandard records, but this notebook does not assume every unmatched row has the same reason.

```
anti_join(closure_school_year, absenteeism_schools, by = c("cds_code", "closure_year")) |>
  select(cds_code, closure_year, county, district_name, school_site, days_wildfire, days_total) |>
  arrange(desc(days_wildfire), county, district_name, school_site) |>
  slice_head(n = 20)
```

A tibble: 20 x 7

	cds_code	closure_year	county	district_name	school_site	days_wildfire
	<chr>	<int>	<chr>	<chr>	<chr>	<dbl>
1	49708620138560	2018	Sonoma	Petaluma Joi~	RISE Acade~	5
2	19101990128025	2018	Los Ange~	Los Angeles ~	Lashon Aca~	3
3	07616550110528	2019	Contra C~	Brentwood Un~	Adams (J. ~	1
4	07616556003644	2019	Contra C~	Brentwood Un~	Brentwood ~	1
5	07616556003651	2019	Contra C~	Brentwood Un~	Edna Hill ~	1
6	07616556070981	2019	Contra C~	Brentwood Un~	Garin Elem~	1
7	07616556117410	2019	Contra C~	Brentwood Un~	Loma Vista~	1
8	07616550108019	2019	Contra C~	Brentwood Un~	Marsh Cree~	1
9	07616550127464	2019	Contra C~	Brentwood Un~	Mary Casey~	1
10	07616550102376	2019	Contra C~	Brentwood Un~	Pioneer El~	1
11	07616556119754	2019	Contra C~	Brentwood Un~	R. Paul Kr~	1
12	07616556109730	2019	Contra C~	Brentwood Un~	Ron Nunn E~	1
13	07616556112718	2019	Contra C~	Brentwood Un~	William B.~	1
14	01100176106751	2018	Alameda	Alameda Coun~	Alameda Co~	0
15	01100176106751	2019	Alameda	Alameda Coun~	Alameda Co~	0
16	01100170137448	2018	Alameda	Alameda Coun~	Aurum Prep~	0
17	01100170129403	2018	Alameda	Alameda Coun~	Epic Chart~	0
18	01100170129403	2019	Alameda	Alameda Coun~	Epic Chart~	0
19	01100170138867	2018	Alameda	Alameda Coun~	Hayward Co~	0
20	01100170138867	2019	Alameda	Alameda Coun~	Hayward Co~	0

i 1 more variable: days_total <dbl>

```
anti_join(absenteeism_schools, closure_school_year, by = c("cds_code", "closure_year")) |>
  select(cds_code, closure_year, county_name, district_name, school_name, eligible_cumulative) |>
  arrange(county_name, district_name, school_name) |>
  slice_head(n = 20)
```

A tibble: 20 x 6

cds_code	closure_year	county_name	district_name	school_name
----------	--------------	-------------	---------------	-------------

	<chr>	<dbl>	<chr>	<chr>	<chr>
1	017509300000000	2019	Alameda	Dublin Unified	District 0~
2	017510100000000	2018	Alameda	Pleasanton Unified	District 0~
3	017510100000000	2019	Alameda	Pleasanton Unified	District 0~
4	076165500000000	2018	Contra Costa	Brentwood Union Element~	District 0~
5	071007400000001	2019	Contra Costa	Contra Costa County Off~	Nonpublic,~
6	076170500000001	2018	Contra Costa	Knightsen Elementary	Nonpublic,~
7	076175400000000	2018	Contra Costa	Mt. Diablo Unified	District 0~
8	076175400000000	2019	Contra Costa	Mt. Diablo Unified	District 0~
9	096190300000001	2018	El Dorado	Lake Tahoe Unified	Nonpublic,~
10	106212500000001	2019	Fresno	Coalinga-Huron Unified	Nonpublic,~
11	106233100000001	2019	Fresno	Orange Center	Nonpublic,~
12	106236400000001	2018	Fresno	Parlier Unified	Nonpublic,~
13	127551500000000	2018	Humboldt	Eureka City Schools	District 0~
14	127551500000000	2019	Humboldt	Eureka City Schools	District 0~
15	156352900000000	2019	Kern	Kern High	District 0~
16	156352900000001	2019	Kern	Kern High	Nonpublic,~
17	156362800000001	2019	Kern	Maricopa Unified	Nonpublic,~
18	156336200000001	2019	Kern	Panama-Buena Vista Union	Nonpublic,~
19	166395800000000	2018	Kings	Kit Carson Union Elemen~	District 0~
20	196421200000000	2018	Los Angeles	ABC Unified	District 0~

i 1 more variable: eligible_cumulative_enrollment <dbl>

Absenteeism by wildfire burden group

```

burden_group_summary <- closure_absenteeism_joined |>
  filter(!is.na(avg_days_absent_per_eligible)) |>
  group_by(burden_group) |>
  summarise(
    school_year_rows = n(),
    schools = n_distinct(cds_code),
    wildfire_days = sum(days_wildfire, na.rm = TRUE),
    wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
    eligible_cumulative_enrollment = sum(eligible_cumulative_enrollment, na.rm = TRUE),
    count_students_one_or_more_absences = sum(count_students_one_or_more_absences, na.rm = TRUE),
    total_days_absent = sum(total_days_absent, na.rm = TRUE),
    share_students_with_absence = safe_divide(
      count_students_one_or_more_absences,
      eligible_cumulative_enrollment
    ),
    avg_days_absent_per_eligible = safe_divide(total_days_absent, eligible_cumulative_enrollment)
  )

```

```

    avg_days_absent_among_absent = safe_divide(
      total_days_absent,
      count_students_one_or_more_absences
    ),
    .groups = "drop"
  )
burden_group_summary

```

```

# A tibble: 3 x 11
  burden_group      school_year_rows schools wildfire_days wildfire_student_days
  <fct>              <int>    <int>         <dbl>             <dbl>
1 No wildfire        16548     9457             0                 0
2 Affected: lower ~    2489     2133            4747            1631238
3 Affected: top 25%     830      780            4888            3650974
# i 6 more variables: eligible_cumulative_enrollment <dbl>,
#   count_students_one_or_more_absences <dbl>, total_days_absent <dbl>,
#   share_students_with_absence <dbl>, avg_days_absent_per_eligible <dbl>,
#   avg_days_absent_among_absent <dbl>

```

```

top_10_cutoff <- closure_absenteeism_joined |>
  filter(wildfire_student_days > 0, !is.na(avg_days_absent_per_eligible)) |>
  summarise(cutoff = quantile(wildfire_student_days, 0.90, na.rm = TRUE)) |>
  pull(cutoff)

if (length(top_10_cutoff) == 0 || is.na(top_10_cutoff)) {
  top_10_cutoff <- Inf
}

make_burden_sensitivity <- function(cutoff, top_label, definition_label) {
  closure_absenteeism_joined |>
    filter(!is.na(avg_days_absent_per_eligible)) |>
    mutate(
      sensitivity_group = case_when(
        wildfire_student_days <= 0 ~ "No wildfire",
        wildfire_student_days >= cutoff ~ top_label,
        TRUE ~ "Affected: below cutoff"
      )
    ) |>
    group_by(sensitivity_group) |>
    summarise(

```

```

    school_year_rows = n(),
    schools = n_distinct(cds_code),
    wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
    eligible_cumulative_enrollment = sum(eligible_cumulative_enrollment, na.rm = TRUE),
    share_students_with_absence = safe_divide(
      sum(count_students_one_or_more_absences, na.rm = TRUE),
      eligible_cumulative_enrollment
    ),
    avg_days_absent_per_eligible = safe_divide(
      sum(total_days_absent, na.rm = TRUE),
      eligible_cumulative_enrollment
    ),
    .groups = "drop"
  ) |>
  mutate(definition = definition_label, .before = 1)
}

burden_group_sensitivity <- bind_rows(
  make_burden_sensitivity(
    affected_cutoff,
    "Affected: top 25%",
    "Top 25% of affected school-years"
  ),
  make_burden_sensitivity(
    top_10_cutoff,
    "Affected: top 10%",
    "Top 10% of affected school-years"
  )
)

burden_group_sensitivity |>
  arrange(definition, sensitivity_group)

```

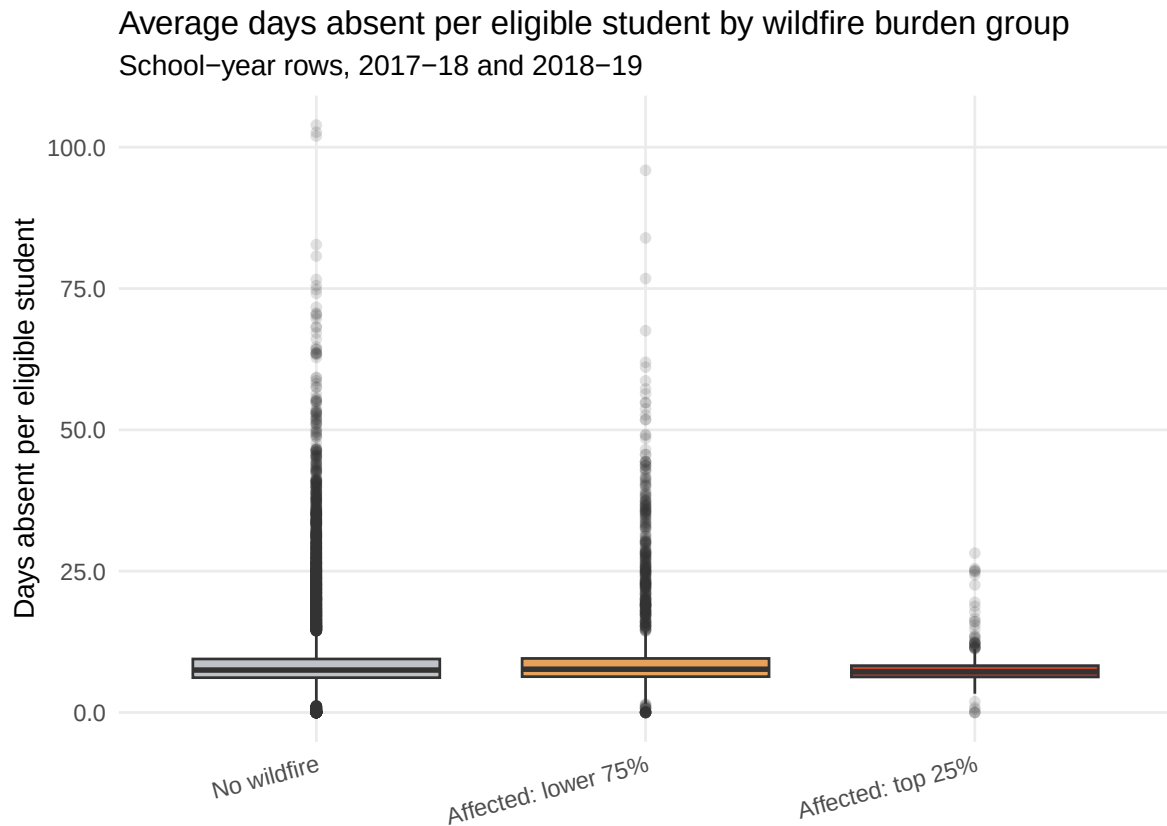
A tibble: 6 x 8

	definition <chr>	sensitivity_group <chr>	school_year_rows <int>	schools <int>	wildfire_student_days <dbl>
1	Top 10% of a~	Affected: below ~	2987	2501	2886488
2	Top 10% of a~	Affected: top 10%	332	322	2395724
3	Top 10% of a~	No wildfire	16548	9457	0
4	Top 25% of a~	Affected: below ~	2489	2133	1631238
5	Top 25% of a~	Affected: top 25%	830	780	3650974
6	Top 25% of a~	No wildfire	16548	9457	0

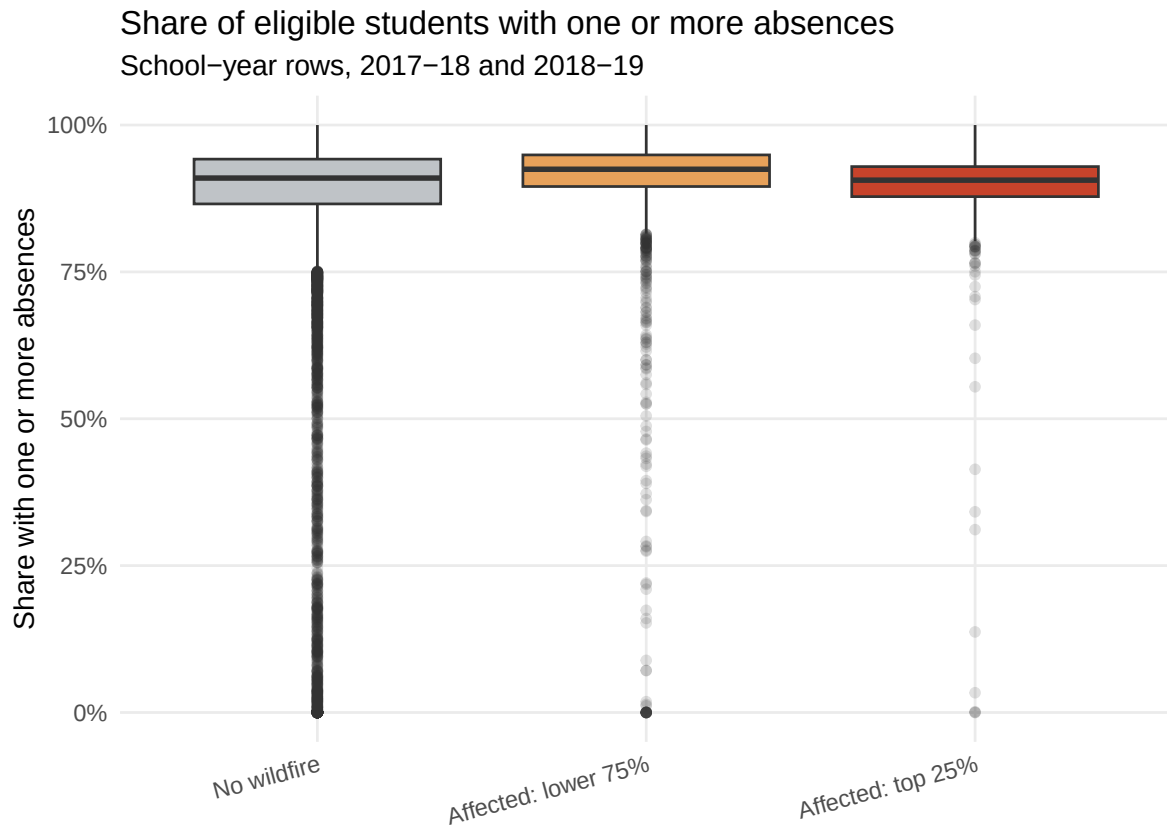
```
# i 3 more variables: eligible_cumulative_enrollment <dbl>,
#   share_students_with_absence <dbl>, avg_days_absent_per_eligible <dbl>
```

The sensitivity table checks whether the descriptive comparison depends entirely on the top-25% cutoff. It should be read cautiously: the highest wildfire-burden group is not automatically expected to have the highest absenteeism value because wildfire burden and absenteeism both vary with school, district, and community context.

```
closure_absenteeism_joined |>
  filter(!is.na(avg_days_absent_per_eligible)) |>
  ggplot(aes(x = burden_group, y = avg_days_absent_per_eligible, fill = burden_group)) +
  geom_boxplot(outlier.alpha = 0.15) +
  scale_fill_manual(values = burden_palette, guide = "none") +
  scale_y_continuous(labels = label_number(accuracy = 0.1)) +
  labs(
    title = "Average days absent per eligible student by wildfire burden group",
    subtitle = "School-year rows, 2017-18 and 2018-19",
    x = NULL,
    y = "Days absent per eligible student"
  ) +
  theme(
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(angle = 15, hjust = 1)
  )
```



```
closure_absenteeism_joined |>
  filter(!is.na(share_students_with_absence)) |>
  ggplot(aes(x = burden_group, y = share_students_with_absence, fill = burden_group)) +
  geom_boxplot(outlier.alpha = 0.15) +
  scale_fill_manual(values = burden_palette, guide = "none") +
  scale_y_continuous(labels = label_percent()) +
  labs(
    title = "Share of eligible students with one or more absences",
    subtitle = "School-year rows, 2017-18 and 2018-19",
    x = NULL,
    y = "Share with one or more absences"
  ) +
  theme(
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(angle = 15, hjust = 1)
  )
```



```
reason_labels <- c(
  "excused_absences_count" = "Excused",
  "unexcused_absences_count" = "Unexcused",
  "out_of_school_suspension_absences_count" = "Suspension",
  "incomplete_independent_study_absences_count" = "Incomplete independent study"
)

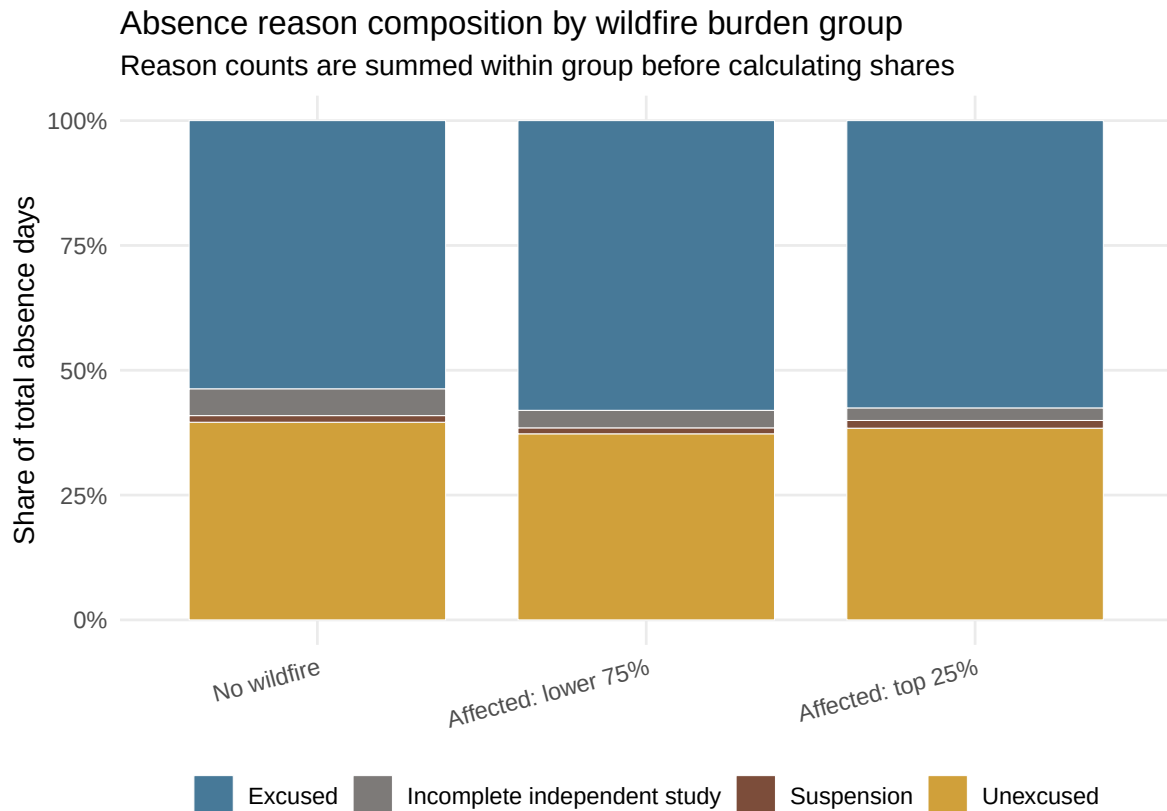
reason_composition_by_group <- closure_absenteeism_joined |>
  filter(!is.na(total_days_absent), total_days_absent > 0) |>
  group_by(burden_group) |>
  summarise(
    total_days_absent = sum(total_days_absent, na.rm = TRUE),
    across(all_of(absence_reason_count_vars), ~ sum(.x, na.rm = TRUE)),
    .groups = "drop"
  ) |>
  pivot_longer(
    cols = all_of(absence_reason_count_vars),
    names_to = "reason",
```

```

    values_to = "absence_days"
  ) |>
  mutate(
    reason = recode(reason, !!!reason_labels),
    reason_share = safe_divide(absence_days, total_days_absent)
  )

ggplot(
  reason_composition_by_group,
  aes(x = burden_group, y = reason_share, fill = reason)
) +
  geom_col(width = 0.78, color = "white", linewidth = 0.2) +
  scale_y_continuous(labels = label_percent()) +
  scale_fill_manual(
    values = c(
      "Excused" = "#477998",
      "Unexcused" = "#D0A03A",
      "Suspension" = "#7C4D3A",
      "Incomplete independent study" = "#7D7A78"
    ),
    name = NULL
  ) +
  labs(
    title = "Absence reason composition by wildfire burden group",
    subtitle = "Reason counts are summed within group before calculating shares",
    x = NULL,
    y = "Share of total absence days"
  ) +
  theme(
    legend.position = "bottom",
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(angle = 15, hjust = 1)
  )

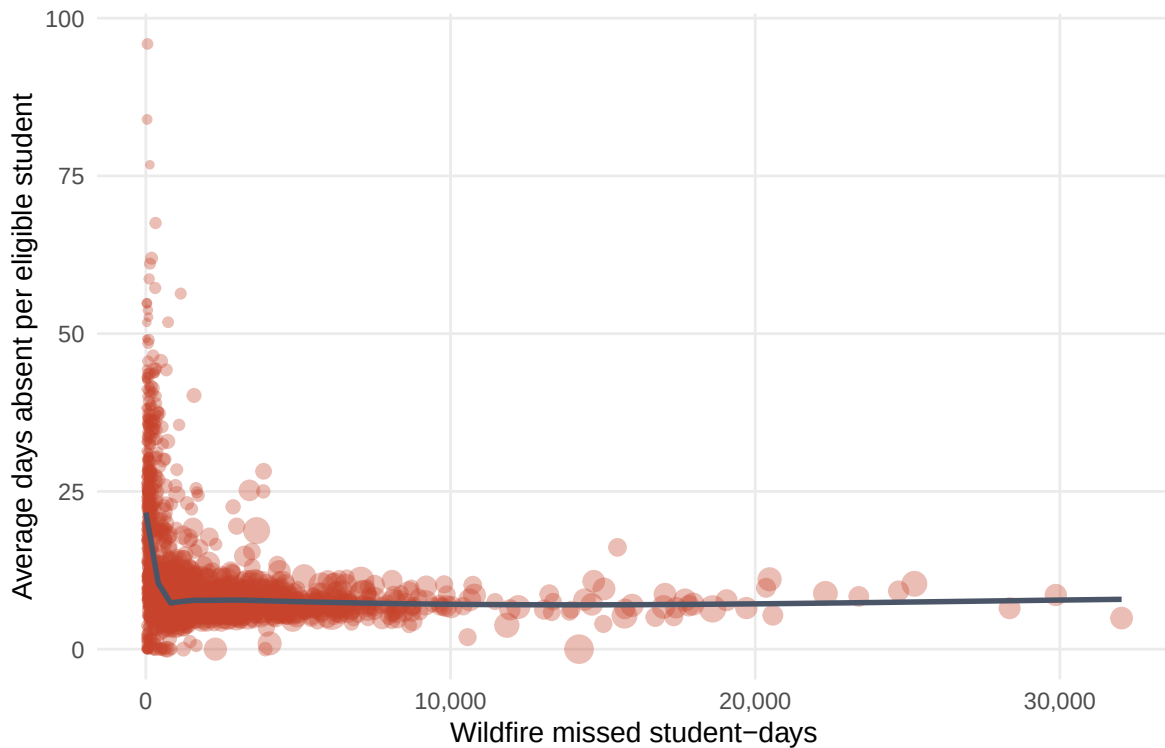
```



```
closure_absenteeism_joined |>
  filter(wildfire_student_days > 0, !is.na(avg_days_absent_per_eligible)) |>
  ggplot(aes(x = wildfire_student_days, y = avg_days_absent_per_eligible)) +
  geom_point(aes(size = eligible_cumulative_enrollment), alpha = 0.35, color = "#C7432B") +
  geom_smooth(method = "loess", se = FALSE, color = "#4A5568") +
  scale_x_continuous(labels = label_comma()) +
  scale_size_continuous(labels = label_comma(), guide = "none") +
  labs(
    title = "Wildfire burden and absenteeism among affected school-years",
    subtitle = "Descriptive only; this is not a causal estimate",
    x = "Wildfire missed student-days",
    y = "Average days absent per eligible student"
  ) +
  theme(panel.grid.minor = element_blank())
```

Wildfire burden and absenteeism among affected school-years

Descriptive only; this is not a causal estimate



County-level absenteeism bridge

```
county_absenteeism_bridge <- closure_absenteeism_joined |>
  filter(!is.na(avg_days_absent_per_eligible), !is.na(county), county != "") |>
  group_by(county) |>
  summarise(
    school_year_rows = n(),
    schools = n_distinct(cds_code),
    wildfire_days = sum(days_wildfire, na.rm = TRUE),
    wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
    eligible_cumulative_enrollment = sum(eligible_cumulative_enrollment, na.rm = TRUE),
    count_students_one_or_more_absences = sum(count_students_one_or_more_absences, na.rm = TRUE),
    total_days_absent = sum(total_days_absent, na.rm = TRUE),
    share_students_with_absence = safe_divide(
      count_students_one_or_more_absences,
      eligible_cumulative_enrollment
    )
  )
```

```

    ),
    avg_days_absent_per_eligible = safe_divide(
      total_days_absent,
      eligible_cumulative_enrollment
    ),
    .groups = "drop"
  ) |>
  mutate(any_wildfire_burden = wildfire_student_days > 0)

county_absenteeism_bridge |>
  arrange(desc(wildfire_student_days)) |>
  select(
    county,
    school_year_rows,
    schools,
    wildfire_days,
    wildfire_student_days,
    eligible_cumulative_enrollment,
    avg_days_absent_per_eligible,
    share_students_with_absence
  ) |>
  slice_head(n = 15)

```

A tibble: 15 x 8

	county	school_year_rows	schools	wildfire_days	wildfire_student_days
	<chr>	<int>	<int>	<dbl>	<dbl>
1	Ventura	414	209	1540	1052437
2	Sonoma	352	180	1768	755293
3	Los Angeles	4445	2252	876	635624
4	Riverside	978	497	366	362906
5	Contra Costa	535	276	584	355246
6	Butte	176	91	907	324435
7	Santa Barbara	239	125	563	271888
8	Napa	84	44	447	219721
9	Sacramento	761	383	304	209112
10	Solano	205	104	251	163045
11	Alameda	763	388	178	129436
12	San Diego	1512	775	166	111195
13	Lake	75	38	339	88738
14	Marin	146	73	214	86485
15	Orange	1219	613	107	78093

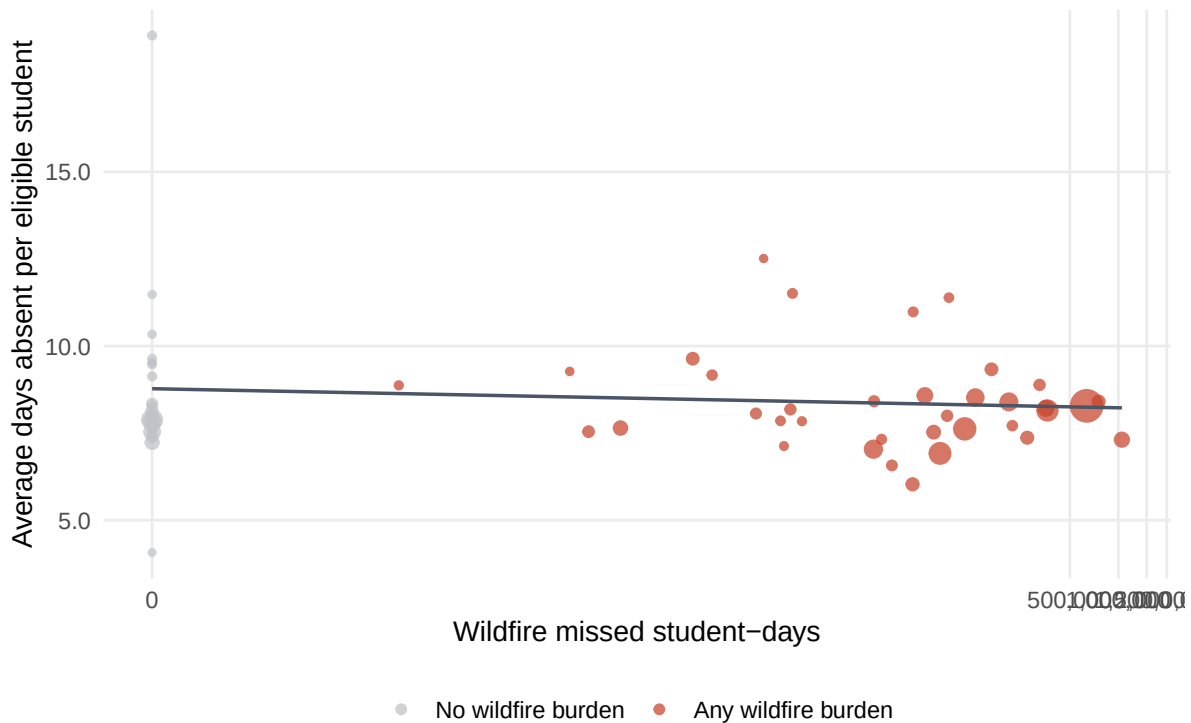
i 3 more variables: eligible_cumulative_enrollment <dbl>,

```
# avg_days_absent_per_eligible <dbl>, share_students_with_absence <dbl>
```

```
ggplot(  
  county_absenteeism_bridge,  
  aes(x = wildfire_student_days, y = avg_days_absent_per_eligible)  
) +  
  geom_point(  
    aes(size = eligible_cumulative_enrollment, color = any_wildfire_burden),  
    alpha = 0.72  
  ) +  
  geom_smooth(method = "lm", se = FALSE, color = "#4A5568", linewidth = 0.7) +  
  scale_x_continuous(  
    trans = pseudo_log_trans(base = 10),  
    labels = label_comma()  
  ) +  
  scale_y_continuous(labels = label_number(accuracy = 0.1)) +  
  scale_size_continuous(labels = label_comma(), guide = "none") +  
  scale_color_manual(  
    values = c("FALSE" = "#BFC3C7", "TRUE" = "#C7432B"),  
    labels = c("No wildfire burden", "Any wildfire burden"),  
    name = NULL  
  ) +  
  labs(  
    title = "County wildfire burden and absenteeism patterns",  
    subtitle = "County summaries aggregate matched school-year rows; descriptive only",  
    x = "Wildfire missed student-days",  
    y = "Average days absent per eligible student"  
  ) +  
  theme(  
    legend.position = "bottom",  
    panel.grid.minor = element_blank()  
  )  
)
```

County wildfire burden and absenteeism patterns

County summaries aggregate matched school-year rows; descriptive only



This county view connects the geographic burden analysis to the absenteeism-by-reason outcomes. It does not estimate the effect of wildfire closures on absenteeism; counties differ in many ways that are not controlled here.

High-burden school profiles

```
school_absenteeism_profile <- closure_absenteeism_joined |>
  group_by(cds_code, county, district_name, school_site) |>
  summarise(
    years_observed = n_distinct(closure_year),
    wildfire_days = sum(days_wildfire, na.rm = TRUE),
    days_total = sum(days_total, na.rm = TRUE),
    wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
    total_enrollment_school_years = sum(total_enrollment, na.rm = TRUE),
    eligible_cumulative_enrollment = sum(eligible_cumulative_enrollment, na.rm = TRUE),
    count_students_one_or_more_absences = sum(count_students_one_or_more_absences, na.rm = TRUE),
    total_days_absent = sum(total_days_absent, na.rm = TRUE),
```

```

    across(all_of(absence_reason_count_vars), ~ sum(.x, na.rm = TRUE)),
    .groups = "drop"
  ) |>
  mutate(
    wildfire_share_of_total = safe_divide(wildfire_days, days_total),
    avg_days_absent_per_eligible = safe_divide(total_days_absent, eligible_cumulative_enrollm
    share_students_with_absence = safe_divide(
      count_students_one_or_more_absences,
      eligible_cumulative_enrollment
    ),
    avg_days_absent_among_absent = safe_divide(
      total_days_absent,
      count_students_one_or_more_absences
    ),
    excused_absence_share = safe_divide(excused_absences_count, total_days_absent),
    unexcused_absence_share = safe_divide(unexcused_absences_count, total_days_absent),
    suspension_absence_share = safe_divide(out_of_school_suspension_absences_count, total_da
    incomplete_independent_study_absence_share = safe_divide(
      incomplete_independent_study_absences_count,
      total_days_absent
    )
  )
)

top_school_absenteeism_profiles <- school_absenteeism_profile |>
  filter(wildfire_student_days > 0) |>
  arrange(desc(wildfire_student_days)) |>
  select(
    county,
    district_name,
    school_site,
    years_observed,
    wildfire_days,
    wildfire_student_days,
    eligible_cumulative_enrollment,
    avg_days_absent_per_eligible,
    share_students_with_absence,
    excused_absence_share,
    unexcused_absence_share,
    suspension_absence_share,
    incomplete_independent_study_absence_share
  ) |>
  slice_head(n = 25)

```

top_school_absenteeism_profiles

A tibble: 25 x 13

	county	district_name	school_site	years_observed	wildfire_days
	<chr>	<chr>	<chr>	<int>	<dbl>
1	Sonoma	Santa Rosa High	Santa Rosa High	2	17
2	Ventura	Ventura Unified	Ventura High	2	15
3	Ventura	Oxnard Union High	Pacifica High	2	10
4	Ventura	Ventura Unified	Buena High	2	15
5	Sonoma	Santa Rosa High	Montgomery High	2	17
6	Ventura	Oxnard Union High	Oxnard High	2	10
7	Sonoma	Santa Rosa High	Maria Carrillo High	2	17
8	Ventura	Oxnard Union High	Channel Islands High	2	10
9	Sonoma	Santa Rosa High	Piner High	2	17
10	Ventura	Oxnard Union High	Adolfo Camarillo High	2	10

i 15 more rows

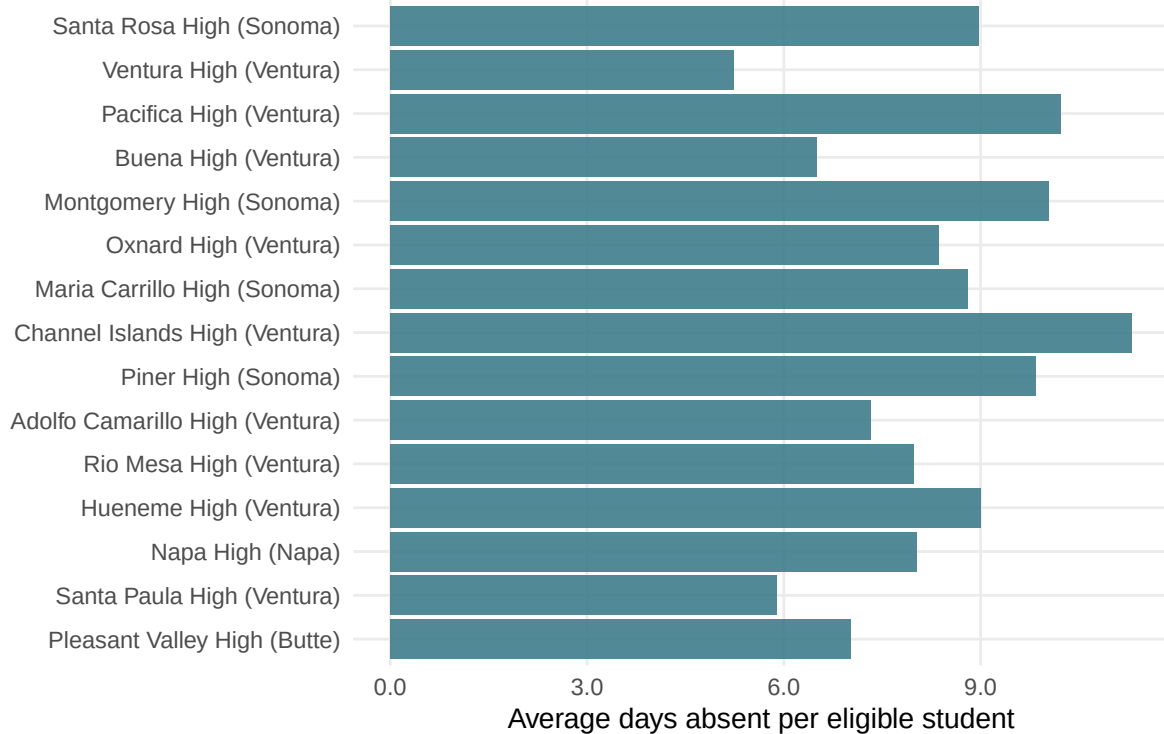
i 8 more variables: wildfire_student_days <dbl>,
eligible_cumulative_enrollment <dbl>, avg_days_absent_per_eligible <dbl>,
share_students_with_absence <dbl>, excused_absence_share <dbl>,
unexcused_absence_share <dbl>, suspension_absence_share <dbl>,
incomplete_independent_study_absence_share <dbl>

```
top_school_plot_data <- school_absenteeism_profile |>
  filter(wildfire_student_days > 0) |>
  slice_max(wildfire_student_days, n = 15, with_ties = FALSE) |>
  mutate(school_label = str_c(str_trunc(school_site, 38), " (", county, ")"))

ggplot(
  top_school_plot_data,
  aes(x = avg_days_absent_per_eligible, y = fct_reorder(school_label, wildfire_student_days))
) +
  geom_col(fill = "#3F7D8C", alpha = 0.9) +
  scale_x_continuous(labels = label_number(accuracy = 0.1)) +
  labs(
    title = "Absenteeism in the highest wildfire-burden schools",
    subtitle = "Top schools ranked by wildfire missed student-days, 2017-18 and 2018-19",
    x = "Average days absent per eligible student",
    y = NULL
  ) +
  theme(panel.grid.minor = element_blank())
```

Absenteeism in the highest wildfire-burden schools

Top schools ranked by wildfire missed student-days, 2017-18 and



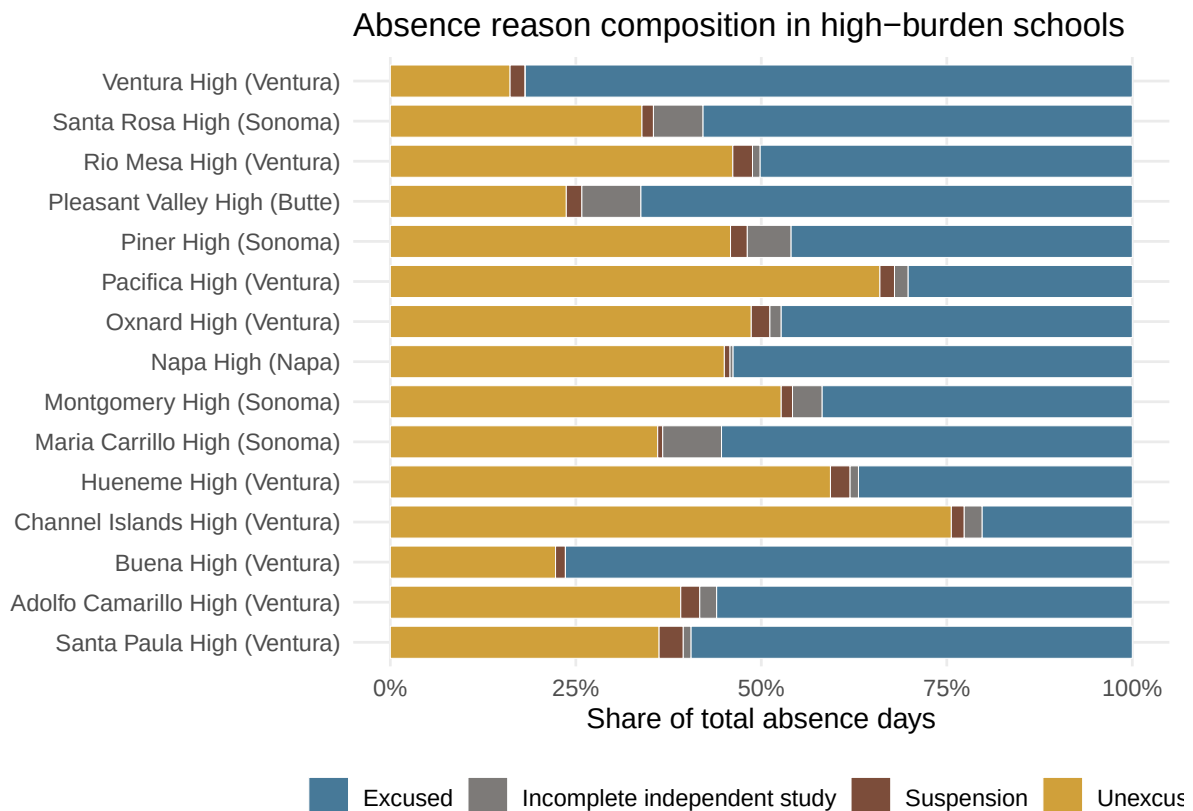
```
top_school_reason_long <- top_school_plot_data |>
  select(school_label, total_days_absent, all_of(absence_reason_count_vars)) |>
  pivot_longer(
    cols = all_of(absence_reason_count_vars),
    names_to = "reason",
    values_to = "absence_days"
  ) |>
  mutate(
    reason = recode(reason, !!!reason_labels),
    reason_share = safe_divide(absence_days, total_days_absent)
  )

ggplot(
  top_school_reason_long,
  aes(x = reason_share, y = fct_reorder(school_label, reason_share, .fun = sum), fill = reason)
) +
  geom_col(width = 0.8, color = "white", linewidth = 0.15) +
  scale_x_continuous(labels = label_percent()) +
```

```

scale_fill_manual(
  values = c(
    "Excused" = "#477998",
    "Unexcused" = "#D0A03A",
    "Suspension" = "#7C4D3A",
    "Incomplete independent study" = "#7D7A78"
  ),
  name = NULL
) +
labs(
  title = "Absence reason composition in high-burden schools",
  x = "Share of total absence days",
  y = NULL
) +
theme(
  legend.position = "bottom",
  panel.grid.minor = element_blank()
)

```



Step 3 takeaway

Matched absenteeism records show descriptive differences across wildfire burden groups and in high-burden school profiles. The pattern is not a causal estimate, and the most wildfire-burdened group is not necessarily the group with the highest average days absent. The safer interpretation is that wildfire burden, absence frequency, and absence reason composition vary together in ways that are worth describing, but the EDA does not isolate wildfire as the cause of absenteeism differences.

Appendix: closure burden and test outcomes

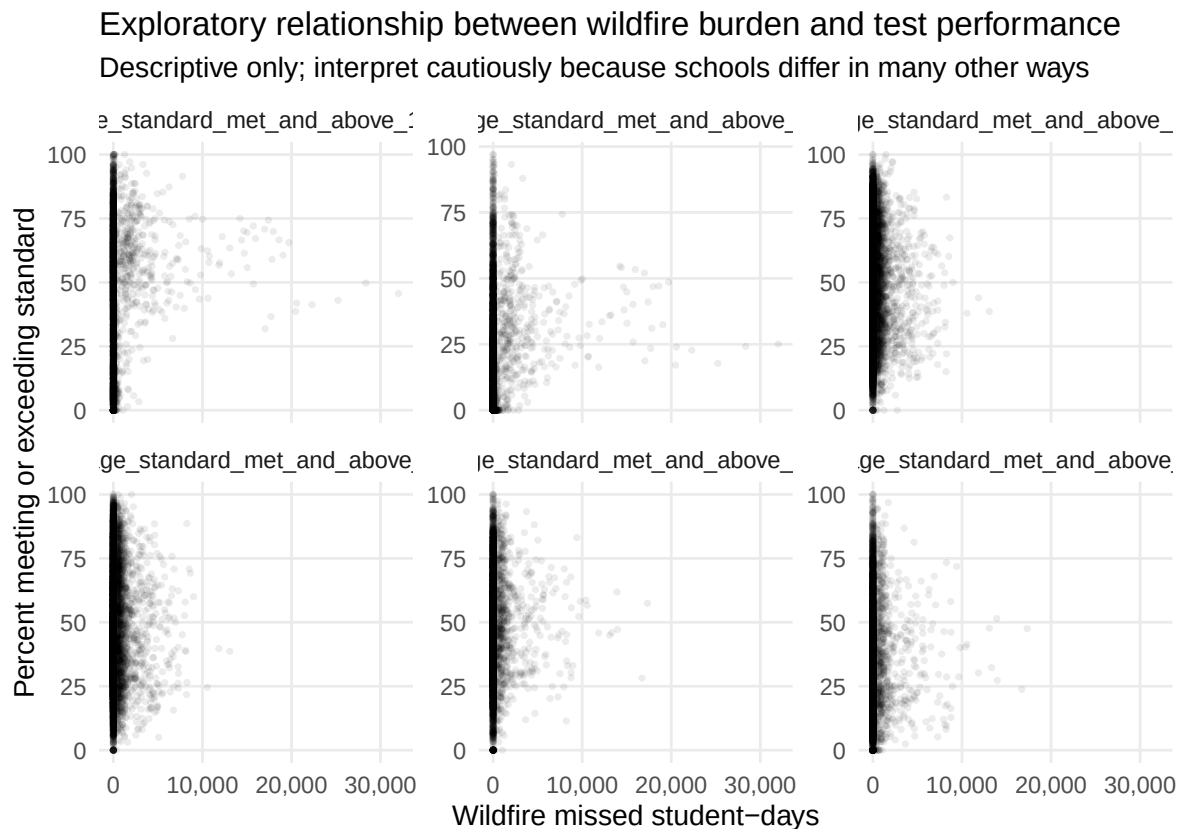
This appendix is retained as an exploratory check only. It is not part of the main wildfire-closure-to-absenteeism story and should not be treated as a causal model.

```
test_score_vars <- c(
  "percentage_standard_met_and_above_3_math",
  "percentage_standard_met_and_above_3_english",
  "percentage_standard_met_and_above_8_math",
  "percentage_standard_met_and_above_8_english",
  "percentage_standard_met_and_above_11_math",
  "percentage_standard_met_and_above_11_english"
)

test_score_long <- df |>
  filter(year %in% analysis_years) |>
  select(cds_code, year, county, days_wildfire, wildfire_student_days, any_of(test_score_vars)) |>
  mutate(across(any_of(test_score_vars), to_number)) |>
  pivot_longer(
    cols = any_of(test_score_vars),
    names_to = "test_metric",
    values_to = "percent_met_or_above"
  ) |>
  filter(!is.na(percent_met_or_above))

ggplot(
  test_score_long,
  aes(x = wildfire_student_days, y = percent_met_or_above)
) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.08, size = 0.7) +
  geom_smooth(method = "loess", se = FALSE, color = "#C7432B") +
  facet_wrap(vars(test_metric), scales = "free_y") +
  scale_x_continuous(labels = label_comma()) +
```

```
labs(
  title = "Exploratory relationship between wildfire burden and test performance",
  subtitle = "Descriptive only; interpret cautiously because schools differ in many other ways",
  x = "Wildfire missed student-days",
  y = "Percent meeting or exceeding standard"
) +
theme(panel.grid.minor = element_blank())
```



Limitations

This EDA is descriptive, not causal. It documents where wildfire closure burden appears in the available closure records and how matched absenteeism-by-reason measures look in those school-years, but it does not estimate what absenteeism would have been without wildfire closures.

The absenteeism analysis is limited to 2017-18 and 2018-19 because those are the CDE Absenteeism by Reason years matched here to closure `year == 2018` and `year == 2019`. The

2019-20 absenteeism file is excluded because CDE marks that year as unavailable/not valid due to COVID-era disruption.

The closure-to-absenteeism match rate is high but incomplete. Unmatched records mean the absenteeism summaries cover most, but not all, closure school-year rows; the unmatched examples should be treated as diagnostics rather than a complete classification of missingness.

Finally, wildfire burden and absenteeism may both reflect other school and community differences, including baseline attendance patterns, grade configuration, district reporting context, socioeconomic conditions, displacement, smoke exposure, local health conditions, and other disruptions. These results are useful for describing patterns and motivating follow-up analysis, not for making causal claims.